MASTER IN ASTROPHYSICS AND SPACE SCIENCE

Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences

Numerical Simulations of AGN Accretion Disks in PLUTO

CAMK PAN Summer Student Program

Fritz Ali Agildere

October 13, 2026

Overview

Overview

Numerical Setup

Overview

Numerical Setup

Black Hole Adaptation

Overview

Numerical Setup

Black Hole Adaptation

Radiation Physics

Overview

Numerical Setup

- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Overview

Numerical Setup

- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier

Overview

Numerical Setup

- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier

- Passive Tracer

Overview

Numerical Setup

- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier

- Passive Tracer
- Construction

Overview

Numerical Setup

- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier

- Passive Tracer
- Construction
- Validation

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results

Overview

Numerical Setup

- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier

- Passive Tracer
- Construction
- Validation

Results

- Relaxation

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State
- Density Waves

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion
- Limitations

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion
- Limitations
- Outlook

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion
- Limitations
- Outlook
- Conclusions

Black Hole Adaptation

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial and Equatorial Symmetry

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial and Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial and Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary
  - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\Phi = -\frac{GM}{r - r_s}$$

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial and Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary
  - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\Phi = -\frac{GM}{r - r_s}$$

  - Measured Lengths and Times in Gravitational Units with c for Velocities

$$r_g = GM/c^2 \qquad t_g = r_g/c$$

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial and Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary
  - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\Phi = -\frac{GM}{r - r_s}$$

  - Measured Lengths and Times in Gravitational Units with c for Velocities

$$r_g = GM/c^2 \qquad t_g = r_g/c$$

  - Embedded Kluźniak–Kita Disk from $R = 6$ with $\epsilon = 0.1$ in Hydrostatic Corona

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial and Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary
  - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\Phi = -\frac{GM}{r - r_s}$$

  - Measured Lengths and Times in Gravitational Units with c for Velocities

$$r_g = GM/c^2 \qquad t_g = r_g/c$$

  - Embedded Kluźniak–Kita Disk from $R = 6$ with $\epsilon = 0.1$ in Hydrostatic Corona
  - Reinjected Analytic Disk at Outer Boundary and Blocked Coronal Inflow

Radiation Physics

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $10^8\,M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\,\mathrm{cm} \qquad t_g \approx 493\,\mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$$

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $10^8\,M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\,\mathrm{cm} \qquad t_g \approx 493\,\mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $10^8\,M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\,\mathrm{cm} \qquad t_g \approx 493\,\mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$
  - Included Kramers Absorption as well as Thomson Scattering Opacities

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $10^8\,M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\,\mathrm{cm} \qquad t_g \approx 493\,\mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$
  - Included Kramers Absorption as well as Thomson Scattering Opacities
- Added Effective Compton Opacity since PLUTO Treats Scattering as Coherent

$$\kappa_C = \kappa_{es} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)}$$

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $10^8\,M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\,\mathrm{cm} \qquad t_g \approx 493\,\mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$
  - Included Kramers Absorption as well as Thomson Scattering Opacities
- Added Effective Compton Opacity since PLUTO Treats Scattering as Coherent

$$\kappa_C = \kappa_{es} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)}$$

  - Reproduces Compton Exchange Rate and Prevents Gas Heating up to 10^{11} K

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $10^8\,M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\,\mathrm{cm} \qquad t_g \approx 493\,\mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$
  - Included Kramers Absorption as well as Thomson Scattering Opacities
- Added Effective Compton Opacity since PLUTO Treats Scattering as Coherent

$$\kappa_C = \kappa_{es} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)}$$

  - Reproduces Compton Exchange Rate and Prevents Gas Heating up to 10^{11} K
  - Rate Capped above 10^9 K and Decoupled above $0.9\,c$ with Gas Floor at 10^4 K

Passive Tracer

Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling, Opacities to Disk

$$\partial_t q + v \cdot \nabla q = 0$$

Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling, Opacities to Disk

$$\partial_t q + v \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves

Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling, Opacities to Disk

$$\partial_t q + v \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves
  - Cause Not Isolated in Code and Only Restricted Physics Affected

Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling, Opacities to Disk

$$\partial_t q + v \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves
  - Cause Not Isolated in Code and Only Restricted Physics Affected
- Replaced Tracer by Classifier Built from Current State of Each Cell Alone

Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling, Opacities to Disk

$$\partial_t q + v \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves
  - Cause Not Isolated in Code and Only Restricted Physics Affected
- Replaced Tracer by Classifier Built from Current State of Each Cell Alone
  - Same Disk Condition as Initial Setup and Identical to Tracer at Start

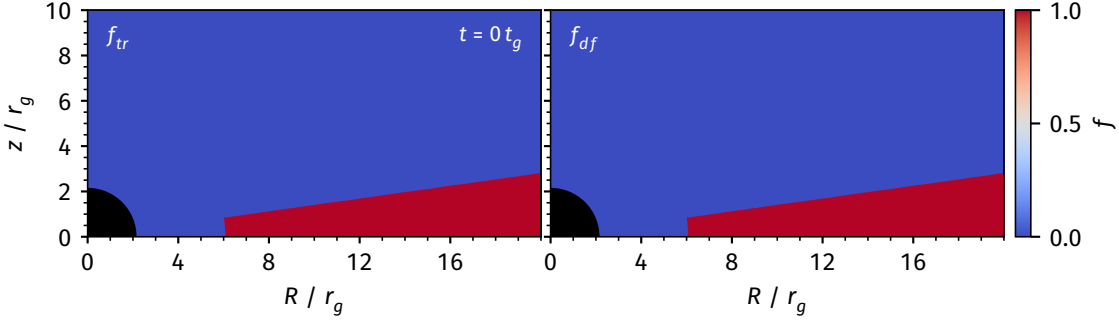
Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling, Opacities to Disk

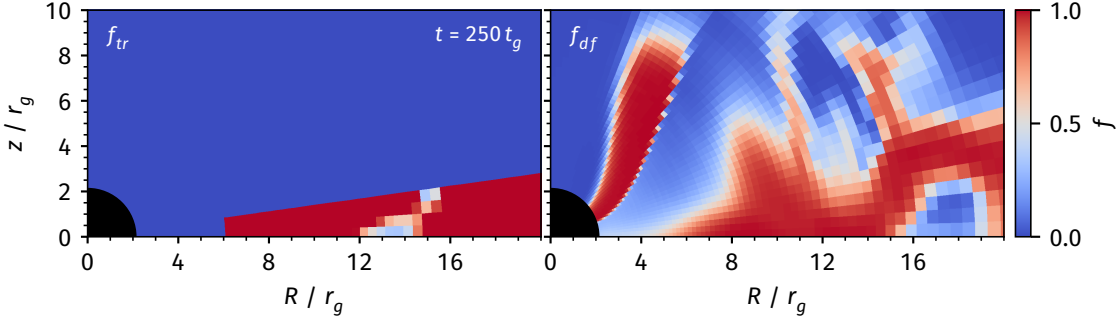
$$\partial_t q + v \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves
  - Cause Not Isolated in Code and Only Restricted Physics Affected
- Replaced Tracer by Classifier Built from Current State of Each Cell Alone
  - Same Disk Condition as Initial Setup and Identical to Tracer at Start
  - Robust to Restarts by Writing Classifier State with Every Output

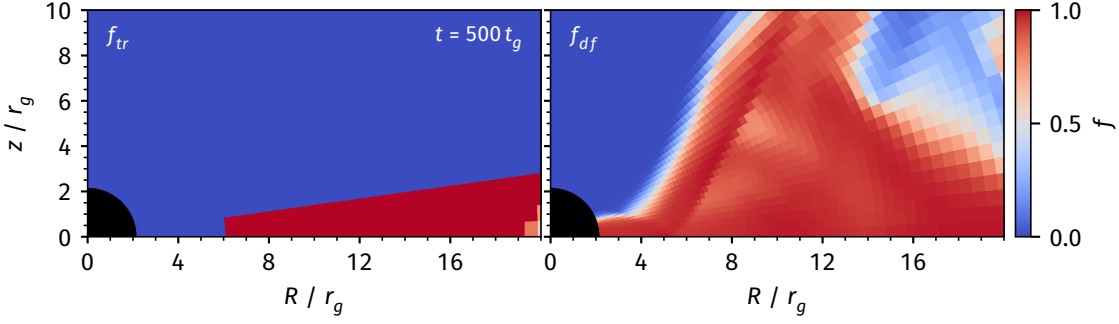
Radiation Enabled – Tracer & Classifier



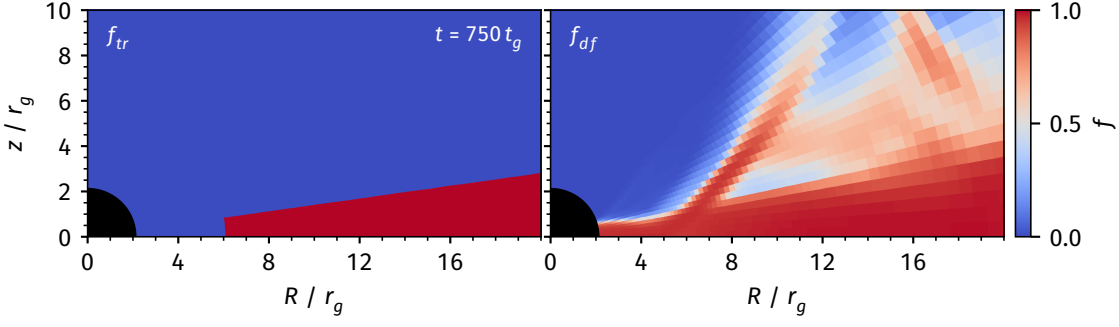
Radiation Enabled – Tracer & Classifier



Radiation Enabled – Tracer & Classifier



Radiation Enabled – Tracer & Classifier



Construction

Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation

Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation
  - Stored Disk and Corona Reference Densities in **90** Logarithmic Radial Bins

Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation
  - Stored Disk and Corona Reference Densities in **90** Logarithmic Radial Bins
  - Located Cell Density between References on Logarithmic Scale

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c, 0.5\right)}$$

Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation
  - Stored Disk and Corona Reference Densities in **90** Logarithmic Radial Bins
  - Located Cell Density between References on Logarithmic Scale

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c, 0.5\right)}$$

  - Multiplied Density and Keplerian Rotation Criteria with Logistic Function

$$f = \sigma\left(\frac{s - 1/6}{0.3}\right)\sigma\left(\frac{|v_\phi|\sqrt{R} - 0.5}{0.1}\right)$$

Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation
  - Stored Disk and Corona Reference Densities in 90 Logarithmic Radial Bins
  - Located Cell Density between References on Logarithmic Scale

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c, 0.5\right)}$$

  - Multiplied Density and Keplerian Rotation Criteria with Logistic Function

$$f = \sigma\left(\frac{s - 1/6}{0.3}\right)\sigma\left(\frac{|v_\phi|\sqrt{R} - 0.5}{0.1}\right)$$

  - Updated References as f and $1 - f$ Weighted Means via Moving Average with Weight $w = 1 - \exp\left(-\Delta t/\tau\right)$ and Time Constant $\tau = 6\, t_g$

Versions & Guards

Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption

Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4

Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification

Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk

Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk
- Current Version Repairs Itself Instead of Locking Error in Place

Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk
- Current Version Repairs Itself Instead of Locking Error in Place
  - Enforced Minimum Span of 0.5 dex Between Disk and Corona References

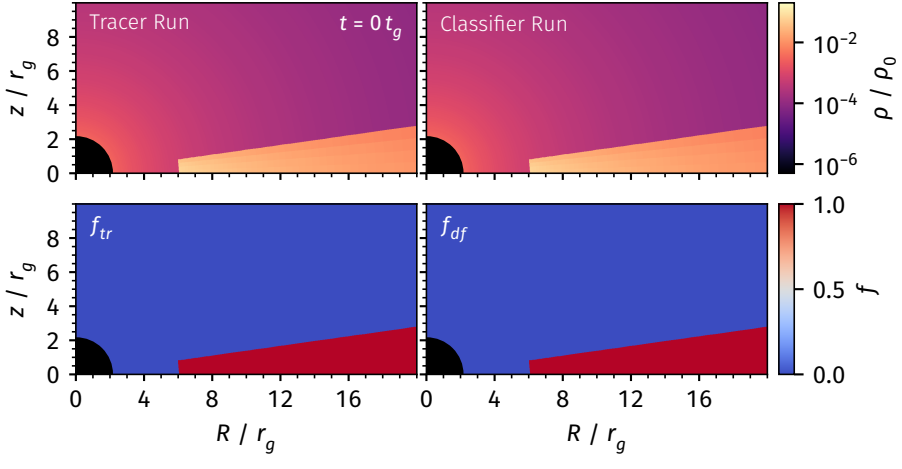
Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk
- Current Version Repairs Itself Instead of Locking Error in Place
  - Enforced Minimum Span of 0.5 dex Between Disk and Corona References
  - Copied References from Nearest Valid Bin with Outward Search First

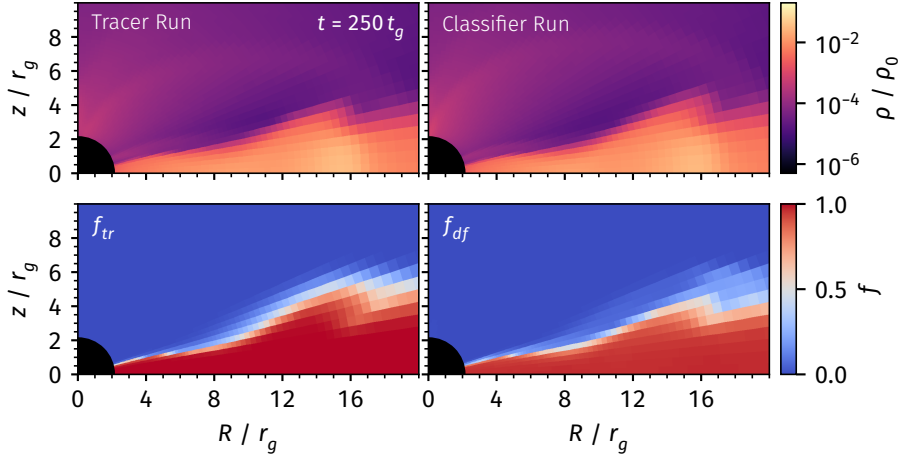
Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk
- Current Version Repairs Itself Instead of Locking Error in Place
  - Enforced Minimum Span of 0.5 dex Between Disk and Corona References
  - Copied References from Nearest Valid Bin with Outward Search First
  - Smoothed with Time Constant for Same Memory at Every Resolution

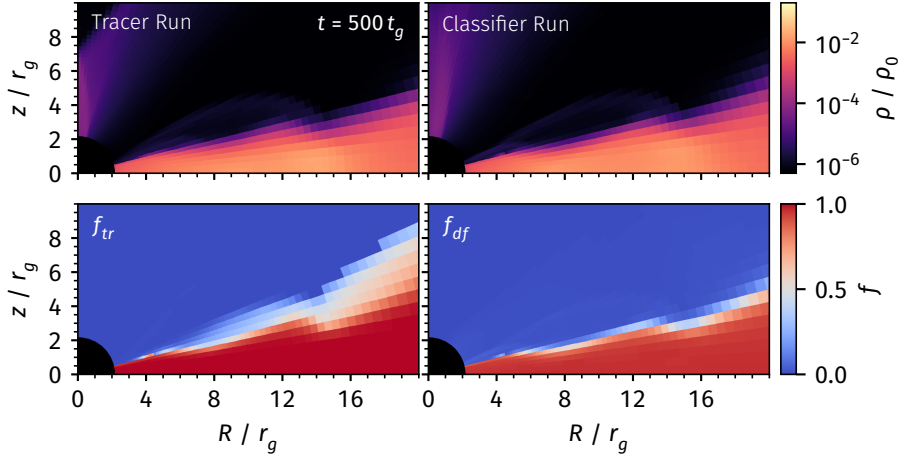
Validation – Hydrodynamics



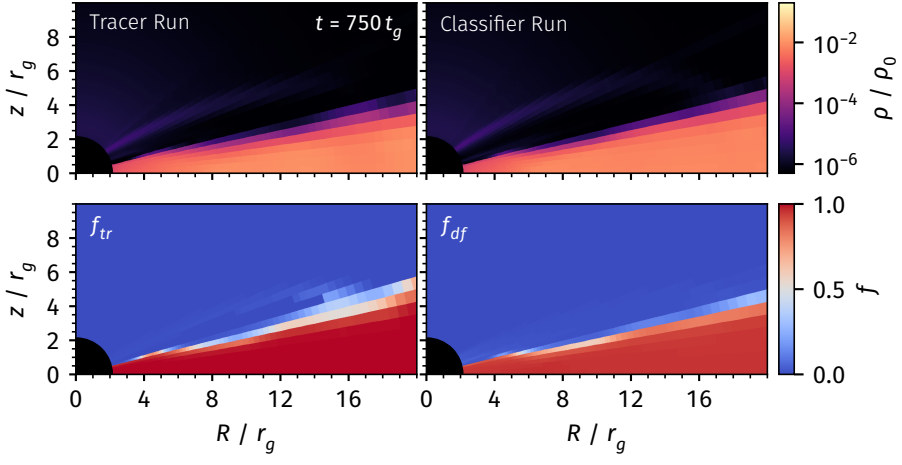
Validation – Hydrodynamics



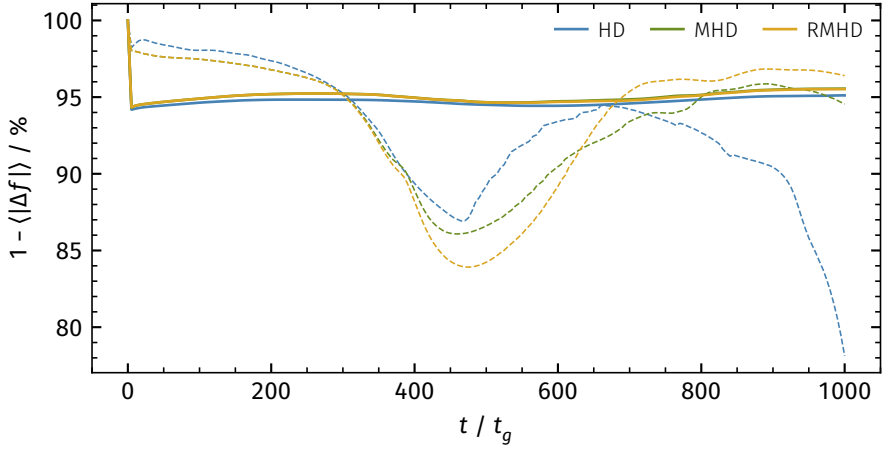
Validation – Hydrodynamics



Validation – Hydrodynamics



Validation – Agreement



Radiation Runs

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with **90 × 45** Cells up to $t = 5 \times 10^4$ and First Classifier Version

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with **90 × 45** Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with **120 × 90** Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every **50** Gravitational Times with Dense Cadence of **5** for Last **10 000**

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times with Dense Cadence of 5 for Last $10\,000$
- Inflated Disk and Breaking Thick Surface Layer during First Hundreds of t_g

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times with Dense Cadence of 5 for Last 10 000
- Inflated Disk and Breaking Thick Surface Layer during First Hundreds of t_g
  - HR Disk Blown Apart and Rebuilt by $t \approx 15000$ while LR Declined Smoothly

Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times with Dense Cadence of 5 for Last 10 000
- Inflated Disk and Breaking Thick Surface Layer during First Hundreds of t_g
  - HR Disk Blown Apart and Rebuilt by $t \approx 15000$ while LR Declined Smoothly
  - Corona Drained Early and Fails to Replenish due to Blocked Inflow

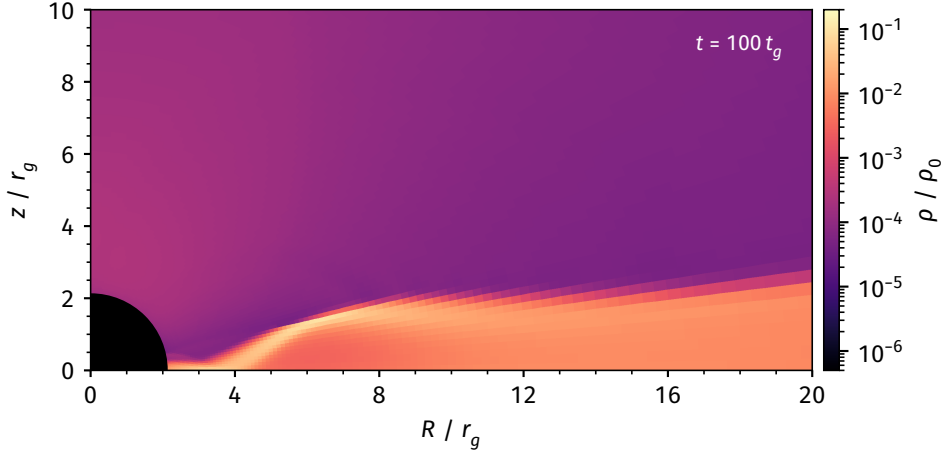
Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times with Dense Cadence of 5 for Last 10 000
- Inflated Disk and Breaking Thick Surface Layer during First Hundreds of t_g
  - HR Disk Blown Apart and Rebuilt by $t \approx 15000$ while LR Declined Smoothly
  - Corona Drained Early and Fails to Replenish due to Blocked Inflow
- Steady State when Disk Mass, Radiation Energy, Accretion Rate, Luminosity Remain within Two Standard Errors of Final Fifth

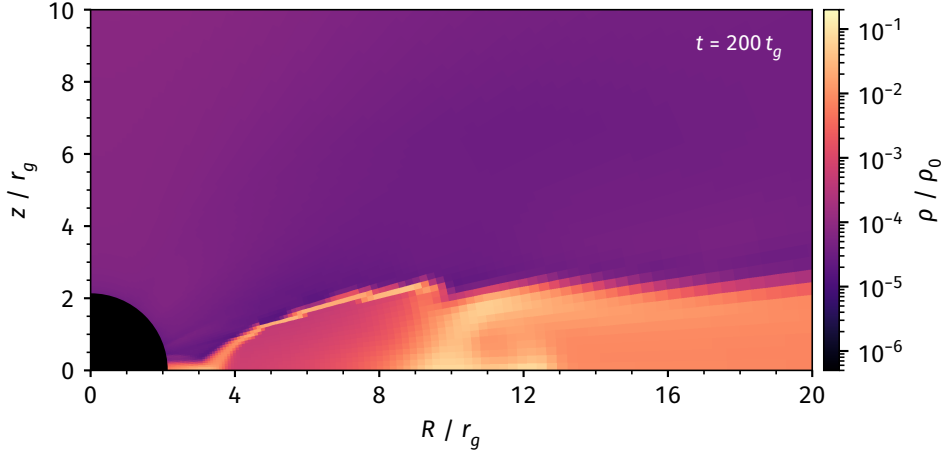
Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times with Dense Cadence of 5 for Last 10 000
- Inflated Disk and Breaking Thick Surface Layer during First Hundreds of t_g
  - HR Disk Blown Apart and Rebuilt by $t \approx 15000$ while LR Declined Smoothly
  - Corona Drained Early and Fails to Replenish due to Blocked Inflow
- Steady State when Disk Mass, Radiation Energy, Accretion Rate, Luminosity Remain within Two Standard Errors of Final Fifth
  - Reached at $t = 23725$ for HR and $t = 9225$ for LR

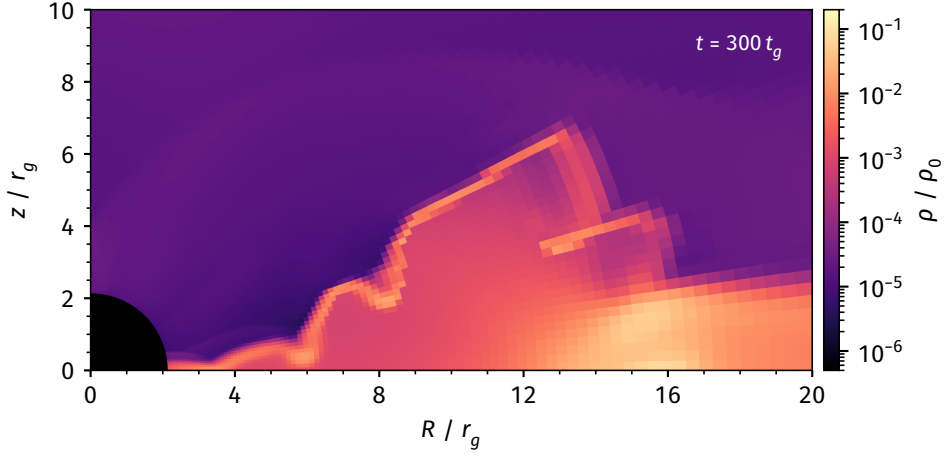
Early Evolution – Density



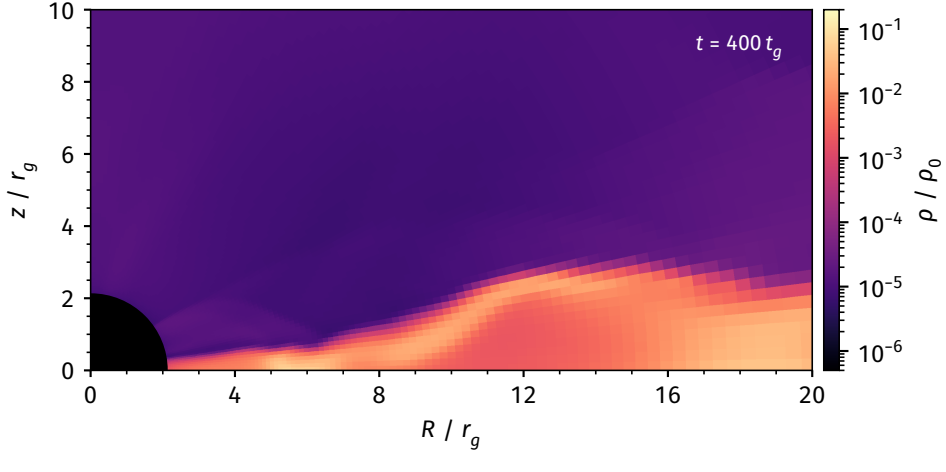
Early Evolution – Density



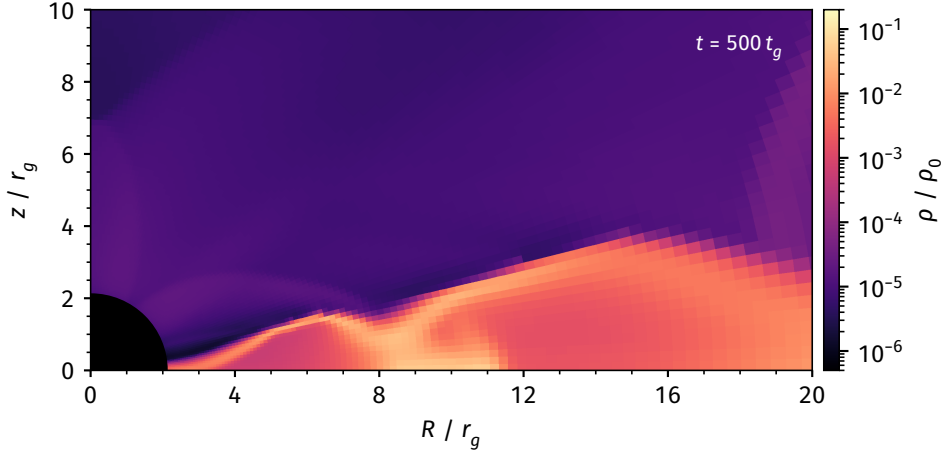
Early Evolution – Density



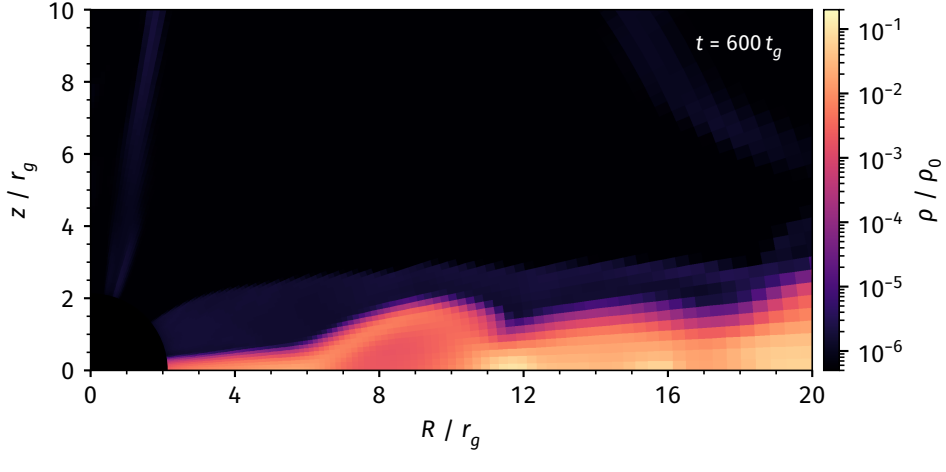
Early Evolution – Density



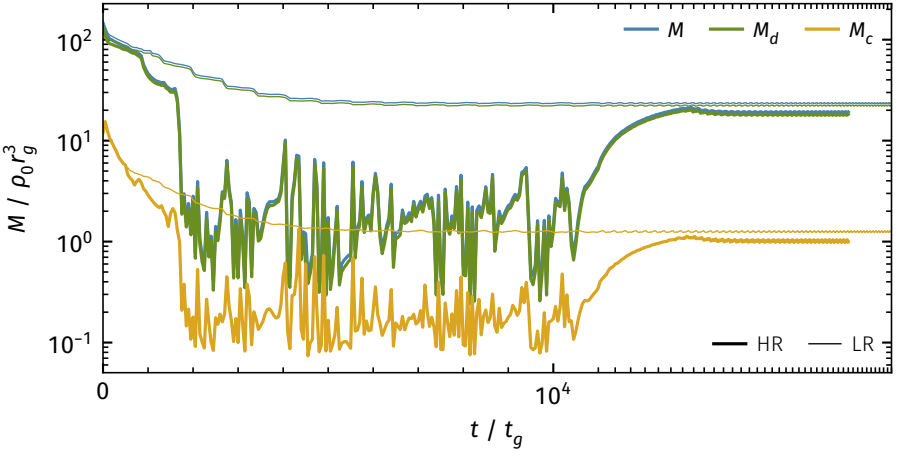
Early Evolution – Density



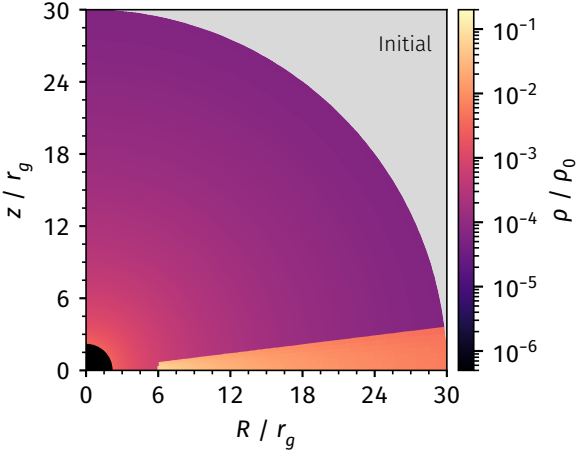
Early Evolution – Density



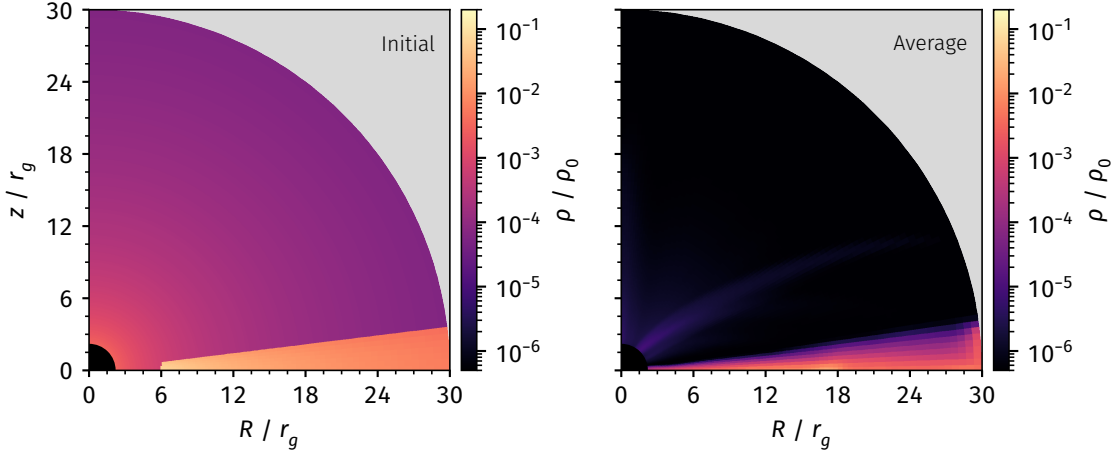
Relaxation – Mass



Steady State – Density



Steady State – Density



Steady State

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond
  - Accretion Rate **0.073** and **0.091** in Eddington Units or about **0.3** $M_\odot\,yr^{-1}$

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond
  - Accretion Rate **0.073** and **0.091** in Eddington Units or about **0.3** $M_\odot\,yr^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond
  - Accretion Rate **0.073** and **0.091** in Eddington Units or about **0.3** $\mathrm{M}_\odot\,\mathrm{yr}^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$
- Surface Density below Steady Thin Disk Profile

$$\Sigma_{ss} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right)$$

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond
  - Accretion Rate **0.073** and **0.091** in Eddington Units or about **0.3** $\mathrm{M}_\odot\,\mathrm{yr}^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$
- Surface Density below Steady Thin Disk Profile

$$\Sigma_{ss} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right)$$

  - Effective Viscosity **2×** to **3×** Higher than Prescribed α

Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond
  - Accretion Rate **0.073** and **0.091** in Eddington Units or about **0.3** $\mathrm{M}_\odot\,\mathrm{yr}^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$
- Surface Density below Steady Thin Disk Profile

$$\Sigma_{ss} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right)$$

  - Effective Viscosity **2×** to **3×** Higher than Prescribed α
- Viscous Heating Balanced by Compton Cooling with Ratio of **0.8** to **0.9**

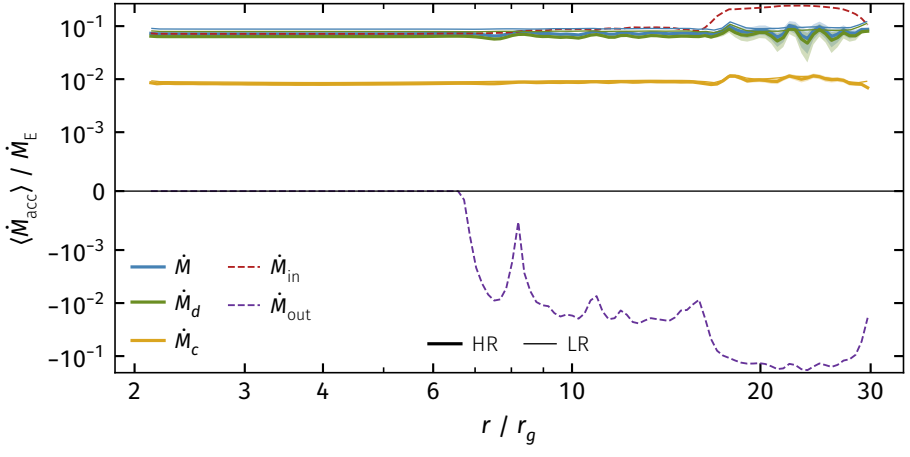
Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Beyond
  - Accretion Rate **0.073** and **0.091** in Eddington Units or about **0.3** $\mathrm{M}_\odot\,\mathrm{yr}^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$
- Surface Density below Steady Thin Disk Profile

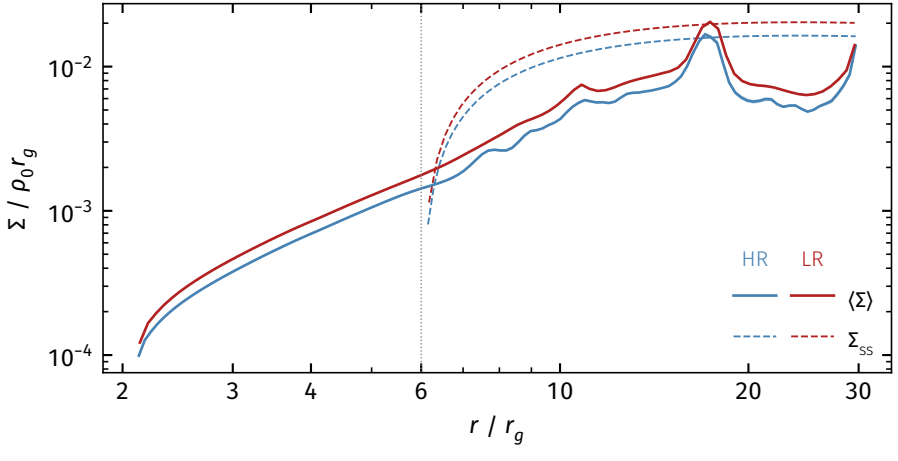
$$\Sigma_{ss} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right)$$

  - Effective Viscosity **2×** to **3×** Higher than Prescribed α
- Viscous Heating Balanced by Compton Cooling with Ratio of **0.8** to **0.9**
  - Gas **100×** to **1000×** Hotter than Radiation at Midplane

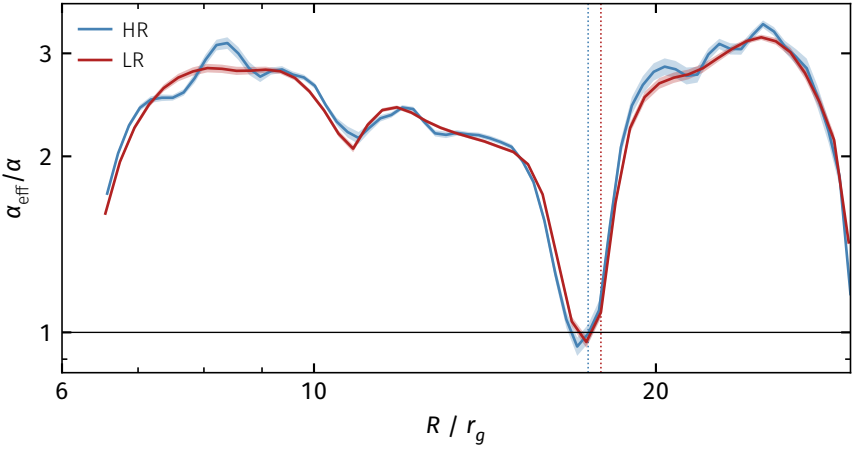
Steady State – Accretion Rate



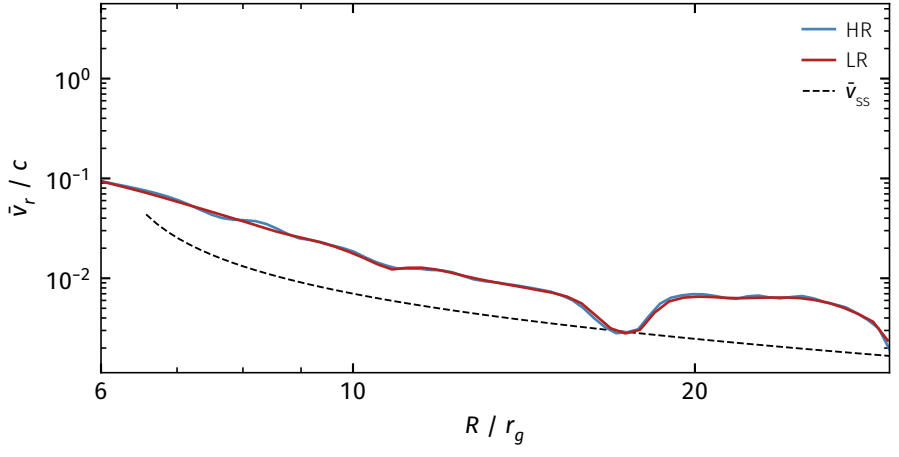
Steady State – Surface Density



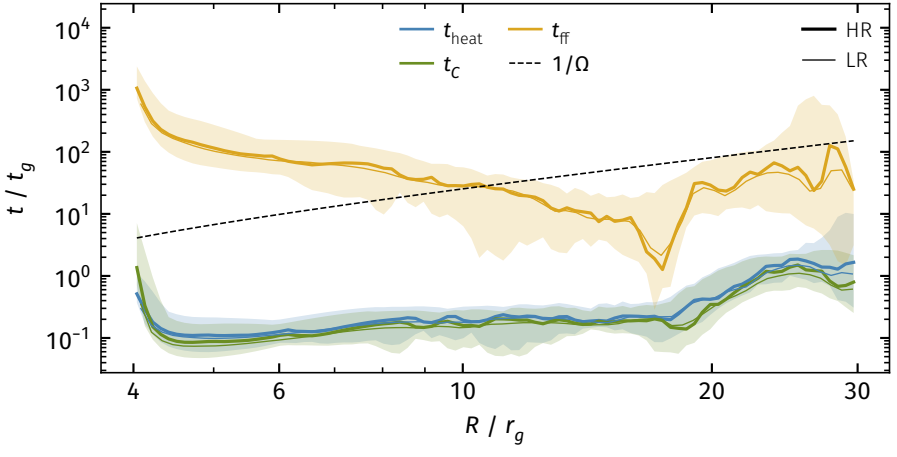
Steady State – Effective Viscosity



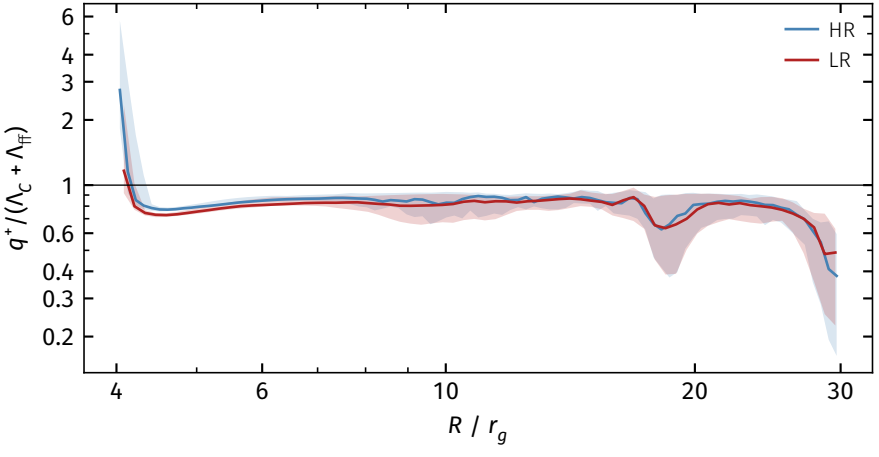
Steady State – Inflow Speed



Steady State – Heating & Cooling Times



Steady State – Heating & Cooling Ratio



Density Waves

Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs

Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond

Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**

Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH

Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH
- Phase Speed of **0.019** c Steady between Radii of **4** and **17**

Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH
- Phase Speed of **0.019** c Steady between Radii of **4** and **17**
  - Roughly Twice Adiabatic Sound Speed with Radiation and Therefore Not Plain Sound Waves

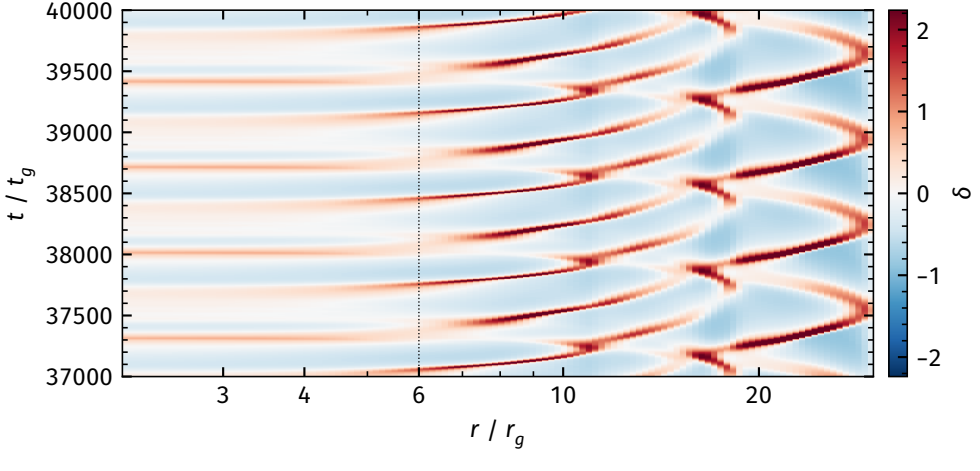
Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH
- Phase Speed of **0.019** c Steady between Radii of **4** and **17**
  - Roughly Twice Adiabatic Sound Speed with Radiation and Therefore Not Plain Sound Waves
  - Same Period at Both Resolutions Suggests Origin in Setup Rather than Numerics

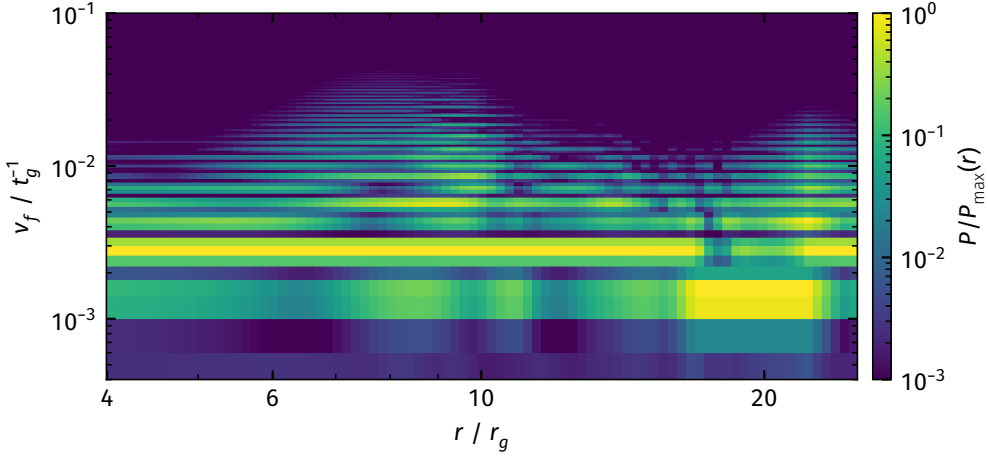
Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH
- Phase Speed of $0.019\,c$ Steady between Radii of **4** and **17**
  - Roughly Twice Adiabatic Sound Speed with Radiation and Therefore Not Plain Sound Waves
  - Same Period at Both Resolutions Suggests Origin in Setup Rather than Numerics
  - Close to Thermal Time $1/(\alpha\Omega)$ of $330\,t_g$ at Outer Boundary

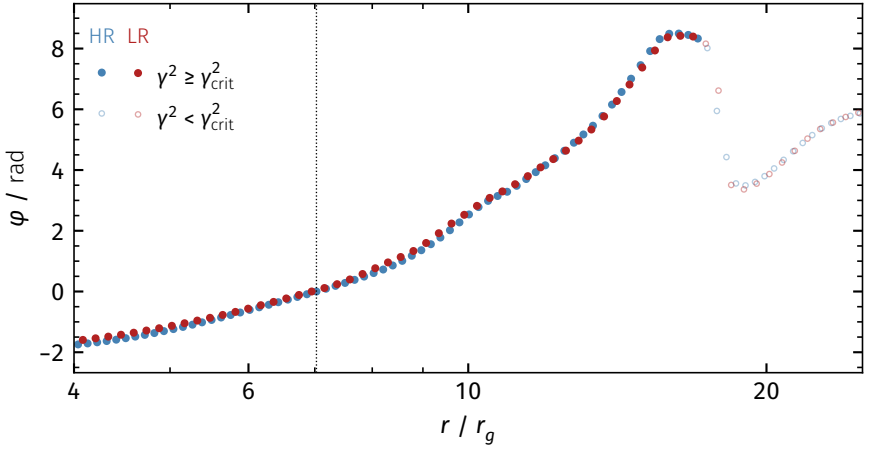
Density Waves – Perturbation



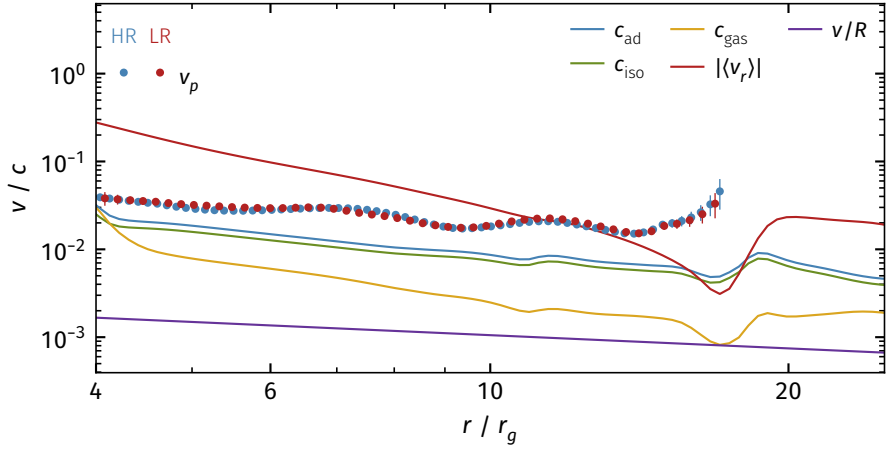
Density Waves – Power Spectrum



Density Waves – Phase Amplitude



Density Waves – Phase Speed



Resolution Comparison

Quantity	HR	LR
Cells ($r \times \theta$)	120 × 90	90 × 45
Classifier Version	Current	First
Steady State Start [t_g]	23 725	9225
Accretion Rate [$\dot{M}_E$]	0.073	0.091
Luminosity [L_E]	0.056	0.070
Efficiency	0.048	0.049
Equilibrium Radius [r_g]	17.4	17.9
Aspect Ratio (R = 10)	0.057	0.042
Cells per Scale Height	3.3	1.2
Wave Period [t_g]	349	349
Phase Speed [c]	0.0190	0.0195

Limitations

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25 with 18 % for Full Domain

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25 with 18 % for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25 with 18 % for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted
  - Corona up to 10^{11} K with $k_B T$ about 46 Times Electron Rest Energy

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period
  - Mass Budget Open by **3 %** between $r = 2.5$ and **25** with **18 %** for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted
  - Corona up to 10^{11} K with $k_B T$ about **46** Times Electron Rest Energy
  - Neither Relativistic Compton Corrections nor Pair Production Included

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25 with 18 % for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted
  - Corona up to 10^{11} K with $k_B T$ about 46 Times Electron Rest Energy
  - Neither Relativistic Compton Corrections nor Pair Production Included
- Plunging Gas inside about Four r_g Moves Superluminally in Pseudo Newtonian Potential

Limitations

- Viscosity Uses Sound Speed of Initial Disk which is Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability Meaning Stability Not Tested
- Boundaries Set Gas Supply which May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25 with 18 % for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted
  - Corona up to 10^{11} K with $k_B T$ about 46 Times Electron Rest Energy
  - Neither Relativistic Compton Corrections nor Pair Production Included
- Plunging Gas inside about Four r_g Moves Superluminally in Pseudo Newtonian Potential
- Axisymmetric Hydrodynamics Replaces MRI Turbulence by α Prescription

Outlook & Conclusions

Outlook & Conclusions

- Next Steps for More Physical Setup

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure
- Built Working SMBH Radiation Setup in PLUTO with Effective Compton Coupling

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure
- Built Working SMBH Radiation Setup in PLUTO with Effective Compton Coupling
  - `DiskFraction` Agrees with Tracer to **95 %** by Mass and Follows Disk with Radiation

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure
- Built Working SMBH Radiation Setup in PLUTO with Effective Compton Coupling
  - `DiskFraction` Agrees with Tracer to **95 %** by Mass and Follows Disk with Radiation
  - Steady States at about a Tenth of Eddington Rate with Compton Balanced Heating

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure
- Built Working SMBH Radiation Setup in PLUTO with Effective Compton Coupling
  - `DiskFraction` Agrees with Tracer to **95 %** by Mass and Follows Disk with Radiation
  - Steady States at about a Tenth of Eddington Rate with Compton Balanced Heating
  - Nonlinear Density Waves with Period of about **350** t_g Traveling Outward

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJS) 2007
- M. Čemeljić, *Atlas of Numerical Solutions for Star-Disk Magnetospheric Interaction* (A&A) 2019
- J. D. Melon Fuksman et al., *A Two Moment Radiation Hydrodynamics Scheme Applicable to Simulations of Planet Formation in Circumstellar Disks* (ApJ) 2021
- B. Paczyński, P. J. Wiita, *Thick Accretion Disks and Supercritical Luminosities* (A&A) 1980
- Y.-F. Jiang et al., *Super Eddington Accretion Disks around Supermassive Black Holes* (ApJ) 2019

Resources on GitHub: here

Model Equations

- Initial Disk Pressure and Density from Analytic Solution of Kluźniak & Kita

$$C = \frac{2}{5\epsilon^2}\left(\frac{1}{r} - \frac{1 - 5\epsilon^2/2}{R}\right) \qquad p = \epsilon^2 C^{5/2} \qquad \rho = C^{3/2}$$

- Corona in Hydrostatic Equilibrium with $\rho_c = 0.01$ in Code Units

$$\rho = \rho_c r^{-3/2} \qquad p = 0.4\rho_c r^{-5/2}$$

- Shear Viscosity with Sound Speed of Initial Disk, Restricted by Classifier

$$\eta_v = \frac{2}{3}\alpha\rho\epsilon^2 C R^{3/2} f \qquad v = \frac{2}{3}\alpha\epsilon^2 R^{1/2}$$

- Density Floor at 5×10^{-7} in Code Units with Constant Sound Speed and Momentum

Opacity Model

- Kramers Free Free and Bound Free Absorption with Guillotine Factor

$$\kappa_K = \left(K_{\rm ff} + K_{\rm bf}\right) \rho\, T_g^{-7/2} \qquad \kappa_{\rm es} = 0.2\,(1 + X)$$

- Compton Exchange Held at Cap above $T_{\rm max} = 1 \times 10^9$ K

$$\kappa_C = \kappa_{\rm es} \frac{4 k_B T_r^4 \left(T_{\rm max} - T_r\right)}{m_e c^2 \left(T_g^4 - T_r^4\right)}$$

- Gated by Classifier and Decoupled near Speed of Light, Thomson Scattering on All Gas

$$f_d = \max\left(\frac{f - 0.05}{0.95}, 0\right) \qquad g_v = \left[1 + \exp\left(\frac{\beta - 0.9}{0.03}\right)\right]^{-1}$$

$$\kappa_{\rm abs} = g_v f_d \left(\kappa_K + \kappa_C\right) \qquad \kappa_{\rm sc} = g_v \left(\kappa_{\rm es} - f_d \kappa_C\right)$$

Classifier Details

- Corona Reference as $1 - f$ and Disk Reference as f Weighted Mean Density per Bin
  - Disk Estimate Counts Only above One Thousandth of Bin Volume
  - Bins without Valid Estimate Copy Nearest Valid Bin
- Guards Added after Inversion in First Version
  - Span below 0.5 dex Copies Both References from Nearest Sufficient Bin
  - Disk Reference Raised to Minimum Span if No Such Bin Exists
  - Fixed Weight 0.05 per Step Replaced by Time Constant τ
- Optional Geometric Mode Independent of Classification
  - Disk Reference from Maximum Density, Corona from Geometric Mean above $z/R = 0.5$

Analysis Methods

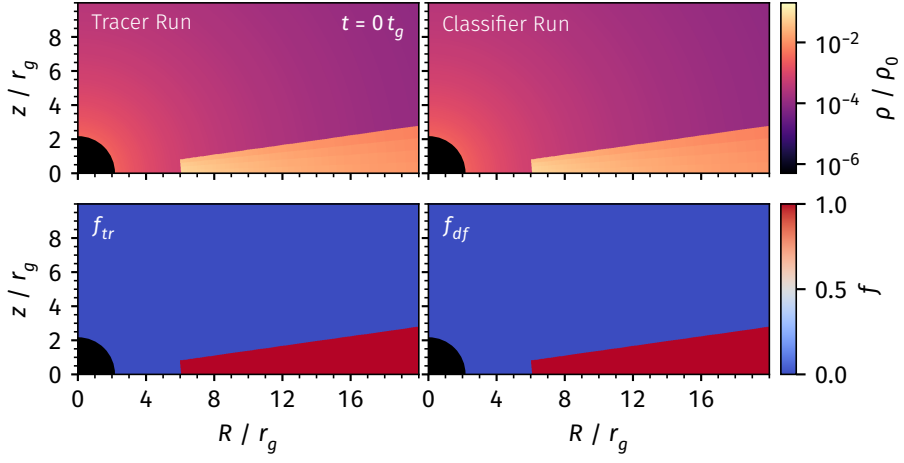
- Standard Errors from Block Averages over **1000** Gravitational Times
- Relaxation from Running Means over $W = 2500$ Compared with Final Fifth

$$D(t_c) = \frac{|\bar{X}_W(t_c) - X_{\mathrm{ref}}|}{\sigma}$$

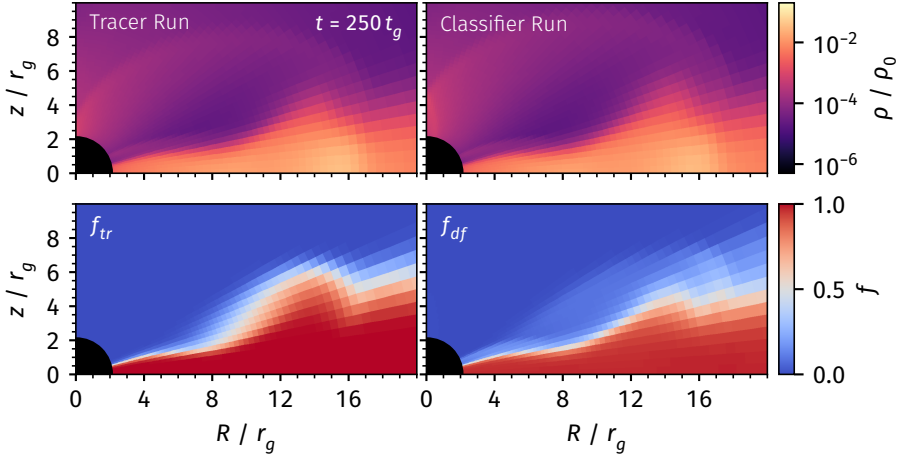
- Inflow Equilibrium where Accretion Rate within **20 %** of Inner Value
- Welch Spectra of Surface Density Perturbation $\delta = \Sigma/\langle\Sigma\rangle - 1$
  - Phase at Peak Frequency Relative to $r = 7$ for Significant Coherence Only
  - Local Phase Speed from Linear Fits over **15 %** in Radius

$$v_p = \omega \,/\, \frac{\mathrm{d}\varphi}{\mathrm{d}r}$$

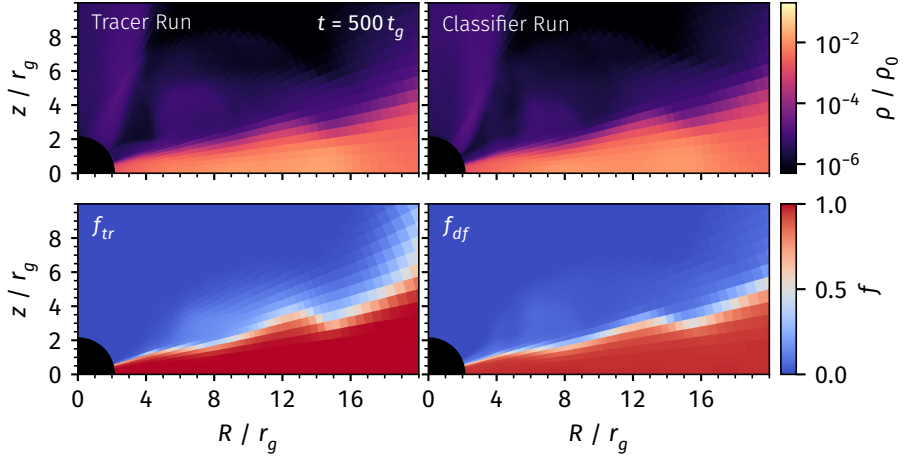
Validation – Magnetohydrodynamics



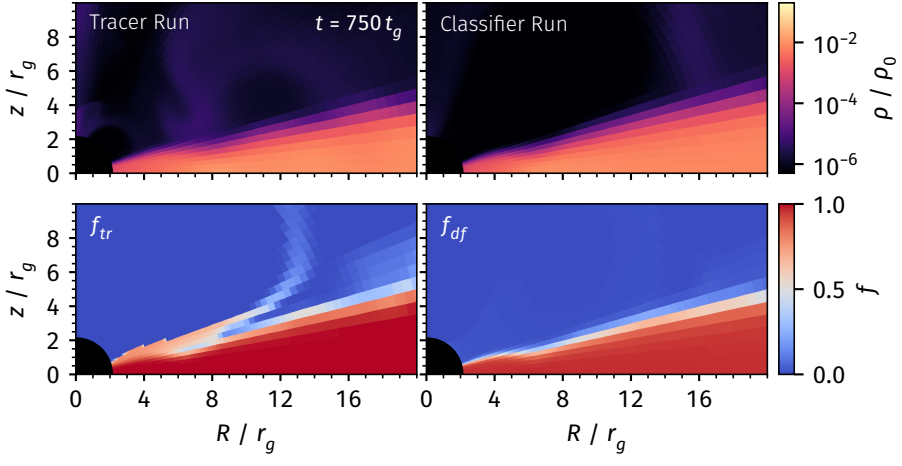
Validation – Magnetohydrodynamics



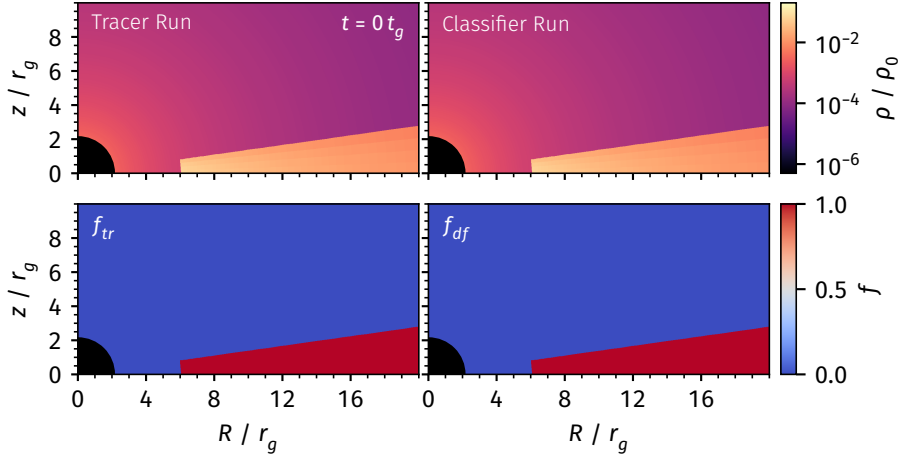
Validation – Magnetohydrodynamics



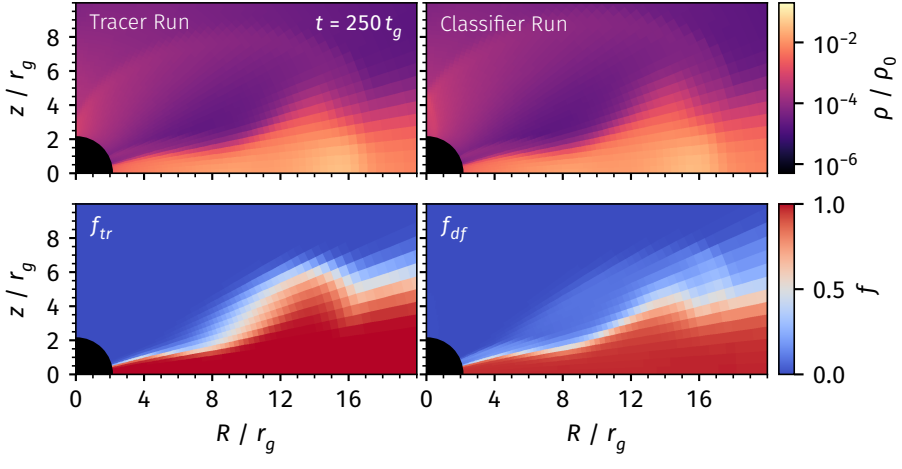
Validation – Magnetohydrodynamics



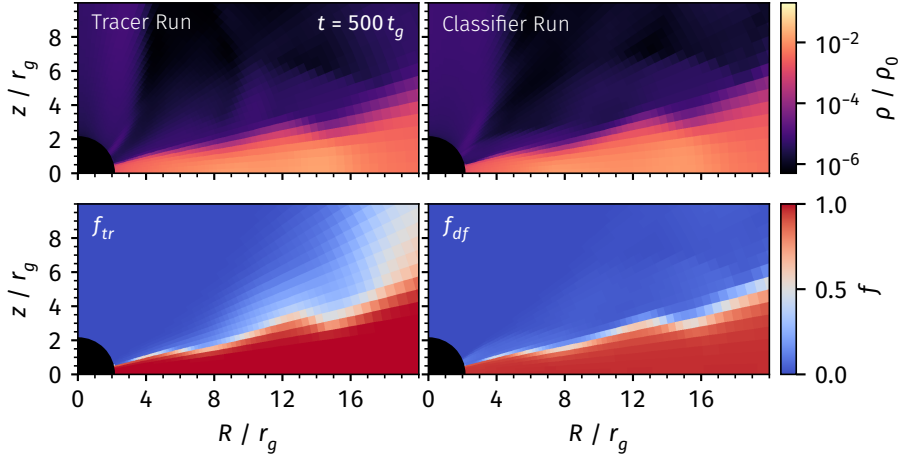
Validation – Resistive Magnetohydrodynamics



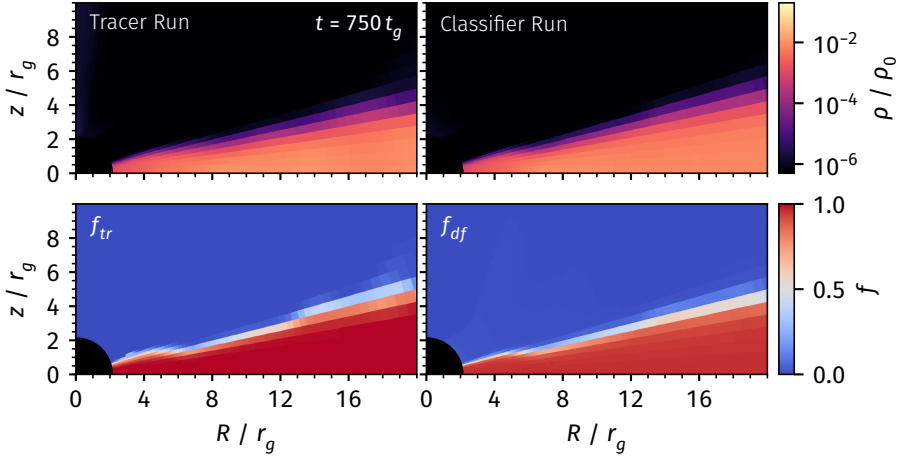
Validation – Resistive Magnetohydrodynamics



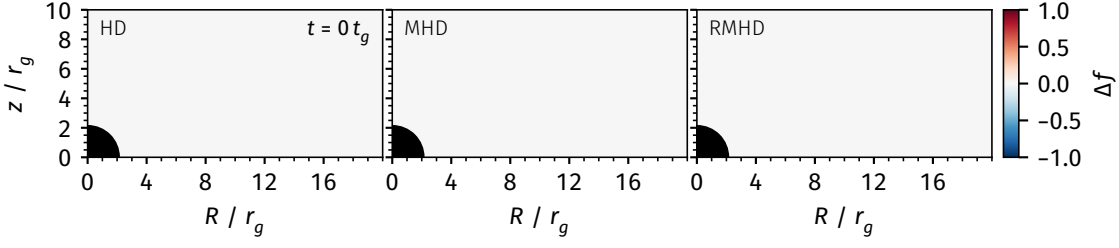
Validation – Resistive Magnetohydrodynamics



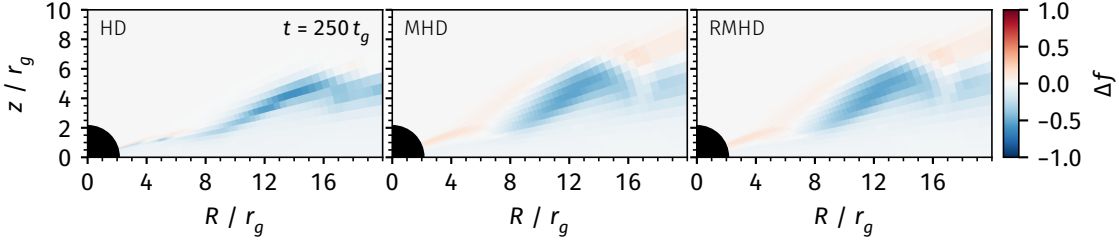
Validation – Resistive Magnetohydrodynamics



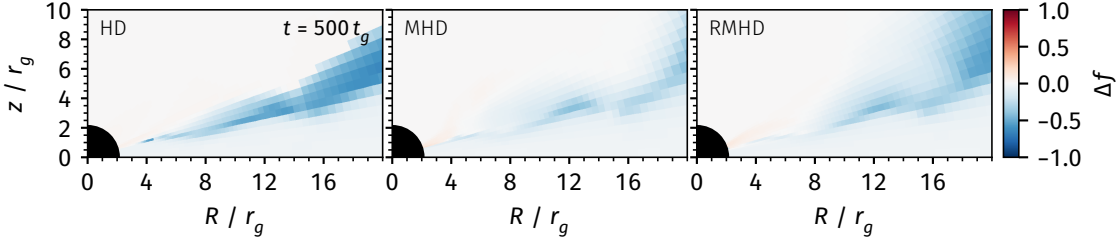
Validation – Classifier Minus Tracer



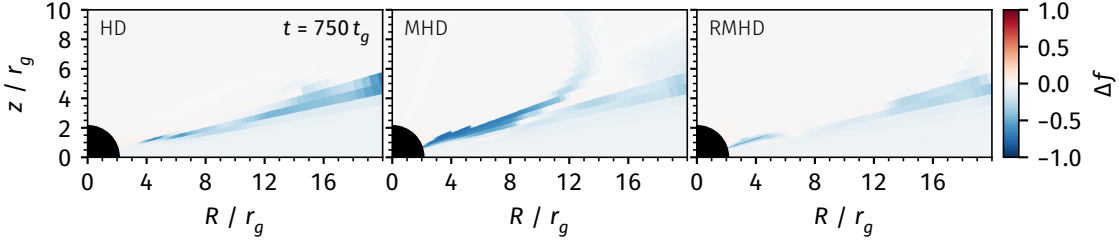
Validation – Classifier Minus Tracer



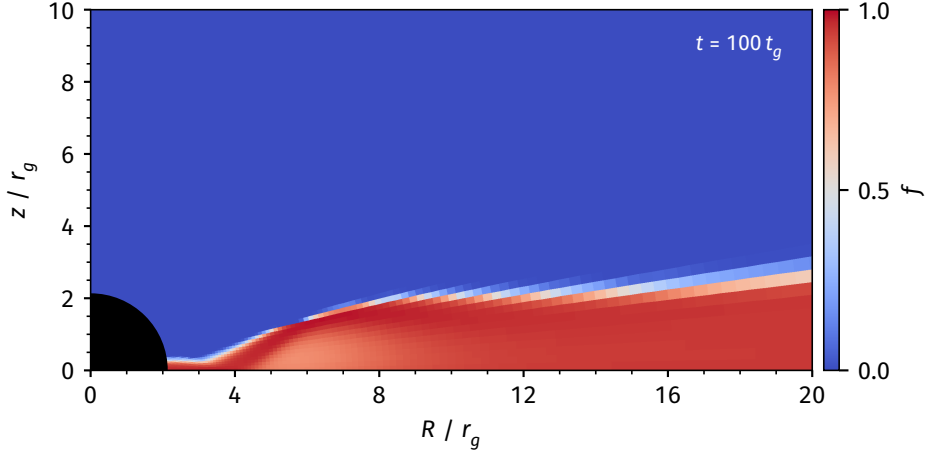
Validation – Classifier Minus Tracer



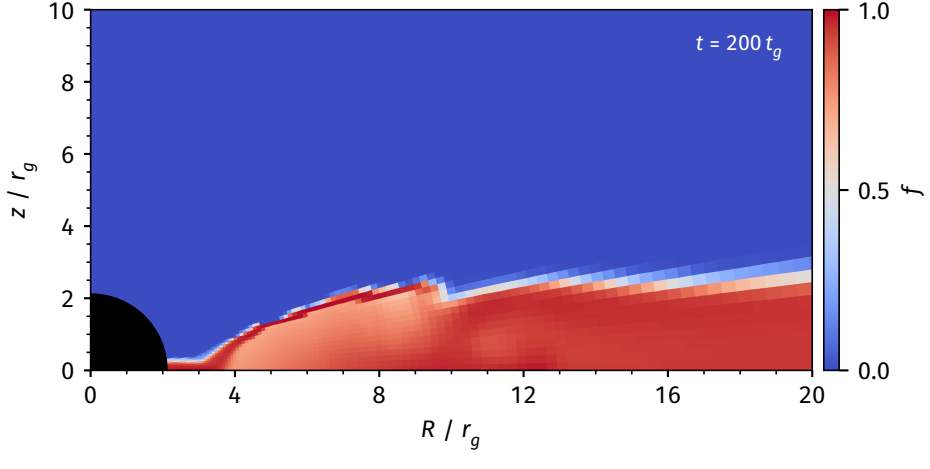
Validation – Classifier Minus Tracer



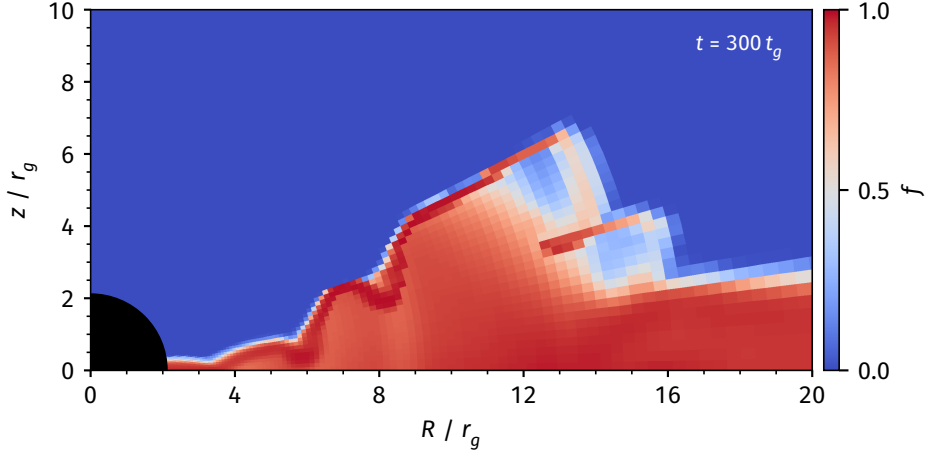
Early Evolution – DiskFraction



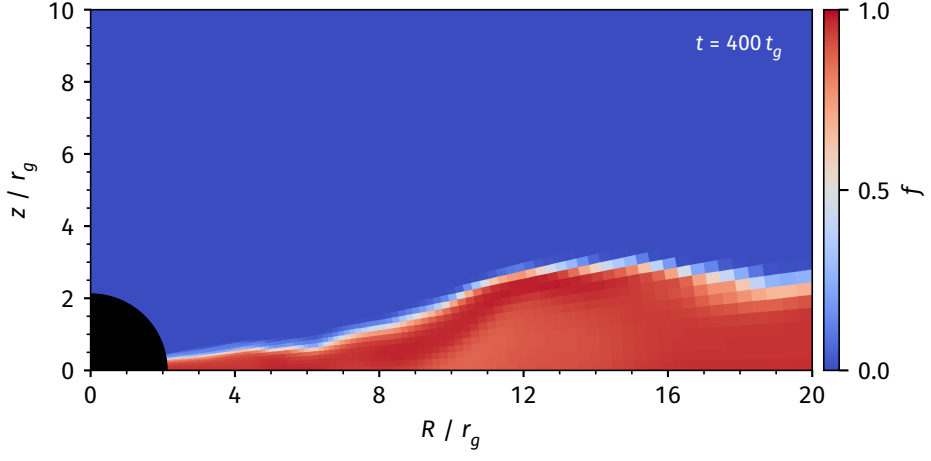
Early Evolution – DiskFraction



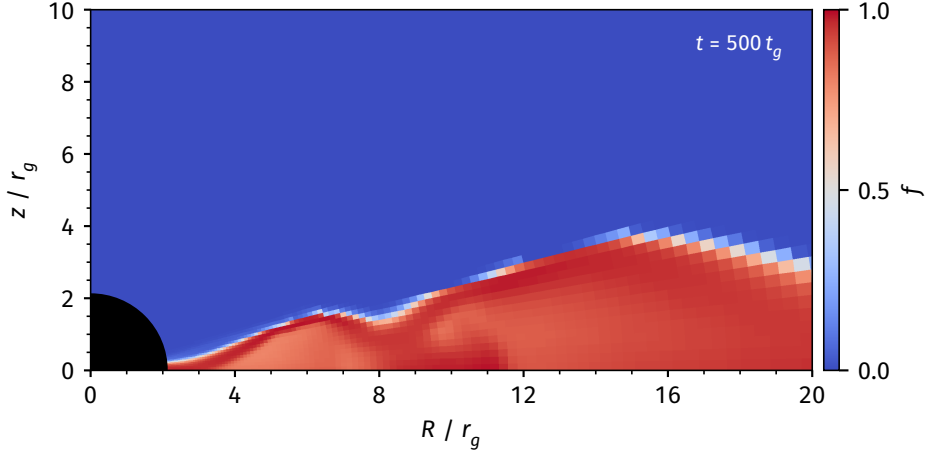
Early Evolution – DiskFraction



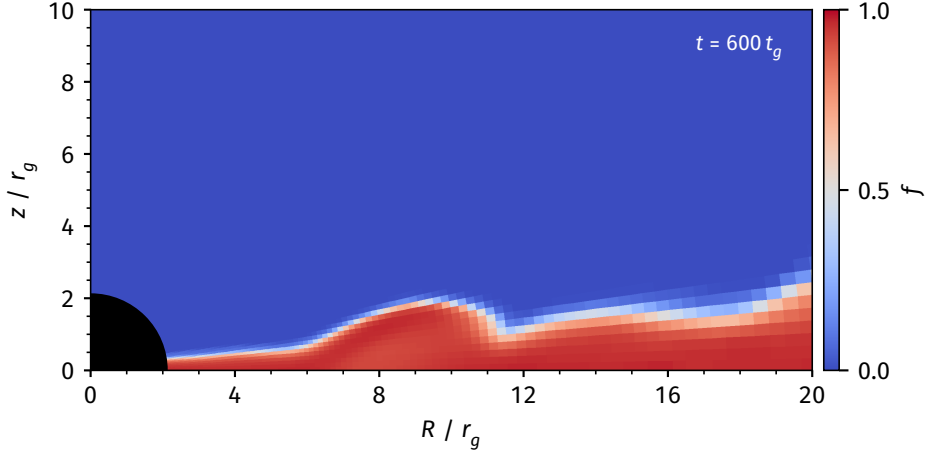
Early Evolution – DiskFraction



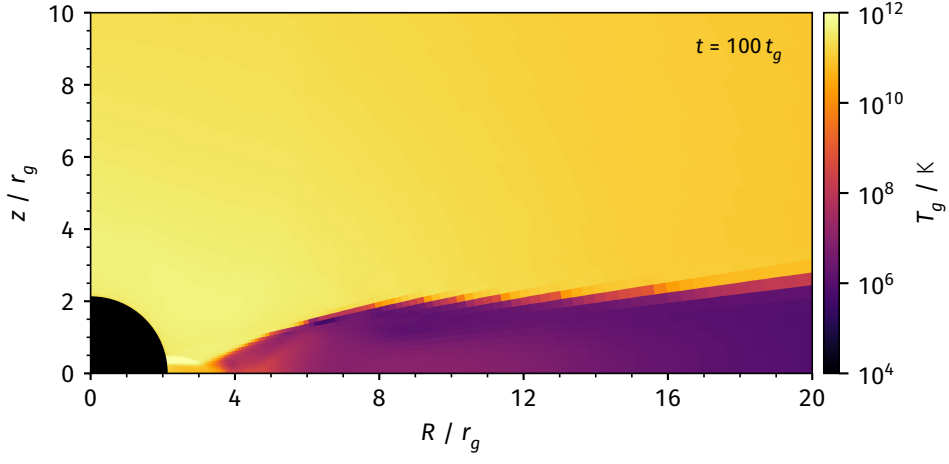
Early Evolution – DiskFraction



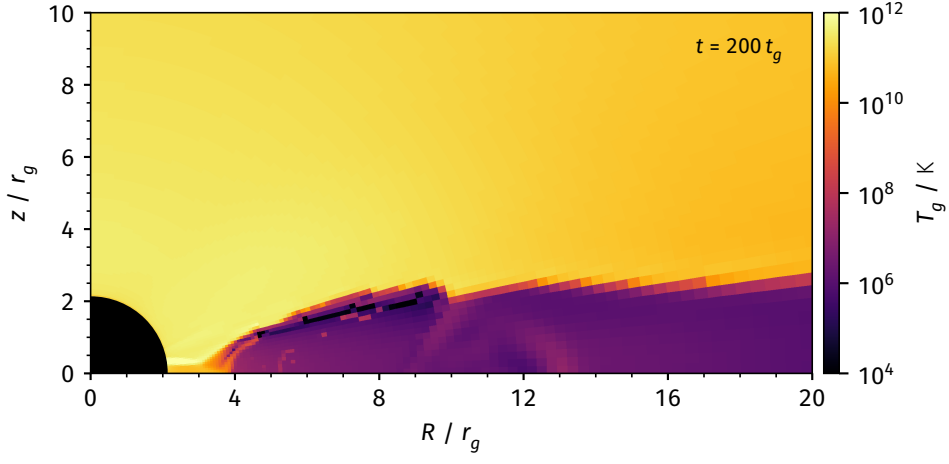
Early Evolution – DiskFraction



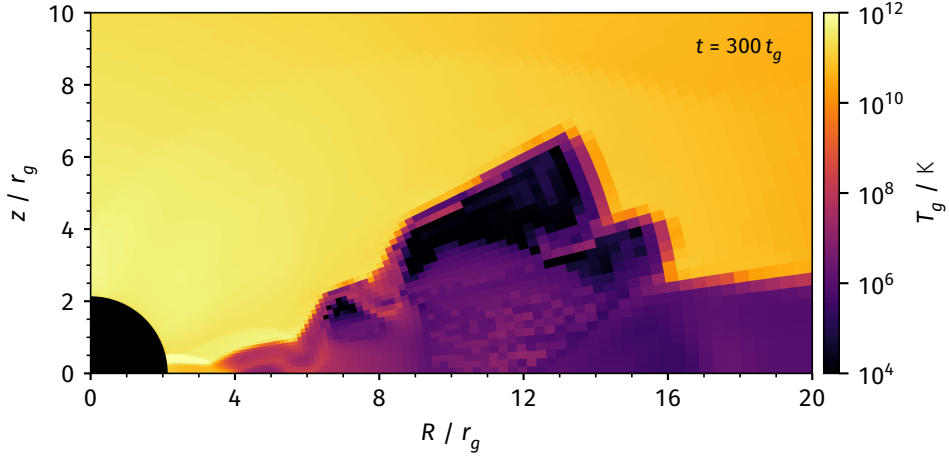
Early Evolution – Gas Temperature



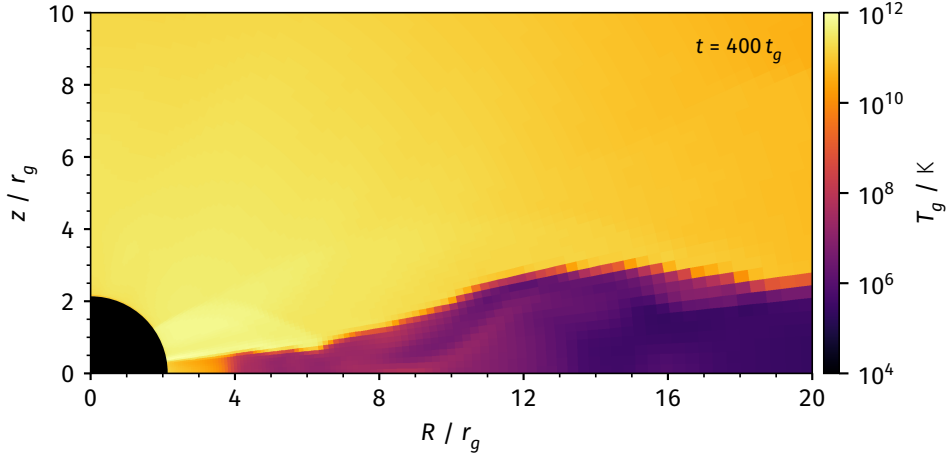
Early Evolution – Gas Temperature



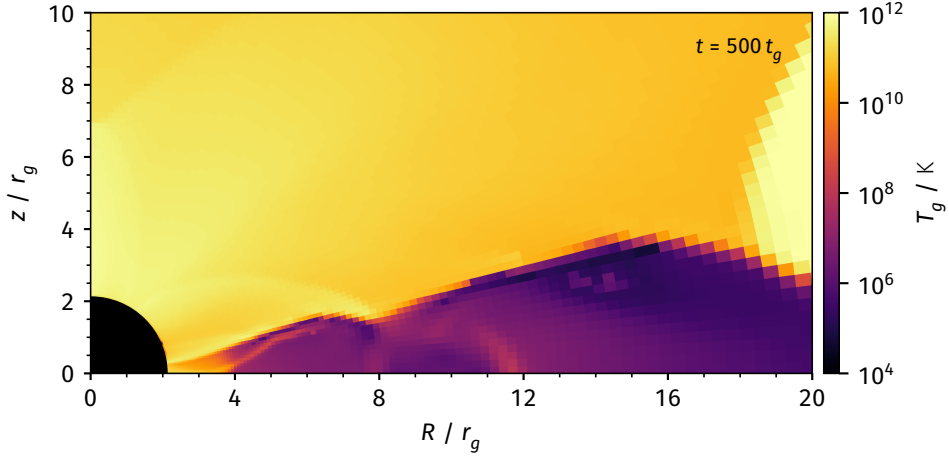
Early Evolution – Gas Temperature



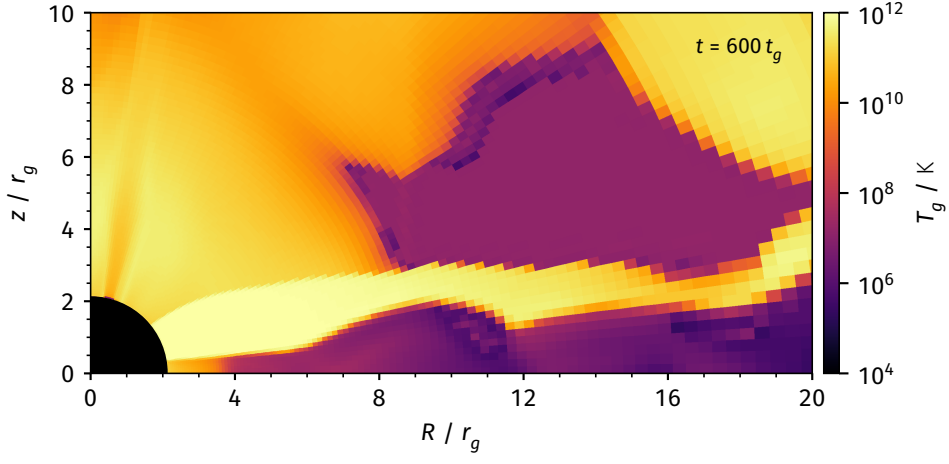
Early Evolution – Gas Temperature



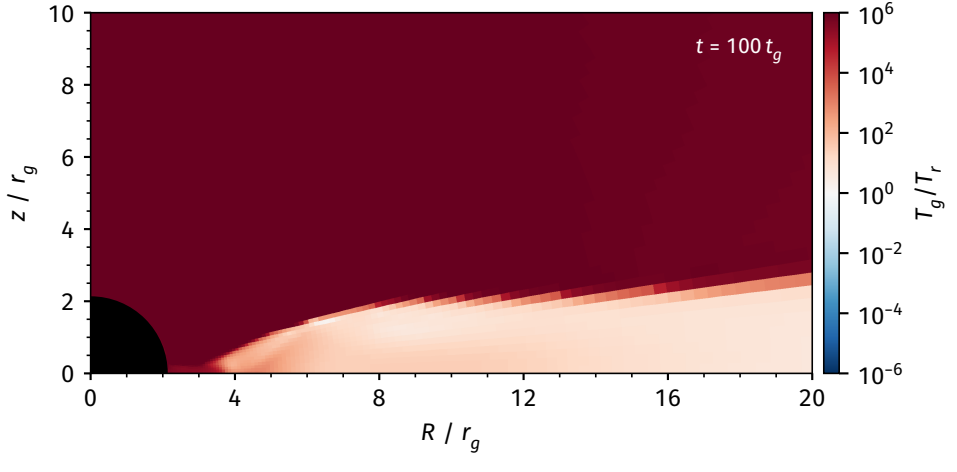
Early Evolution – Gas Temperature



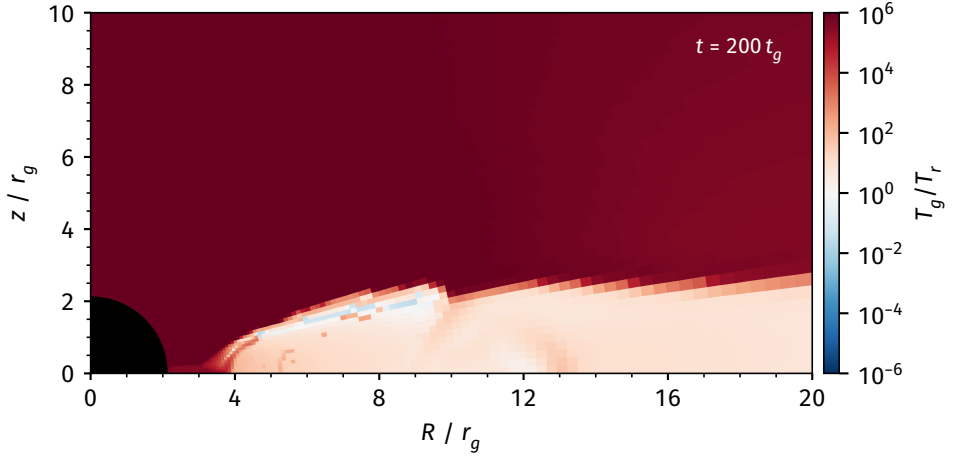
Early Evolution – Gas Temperature



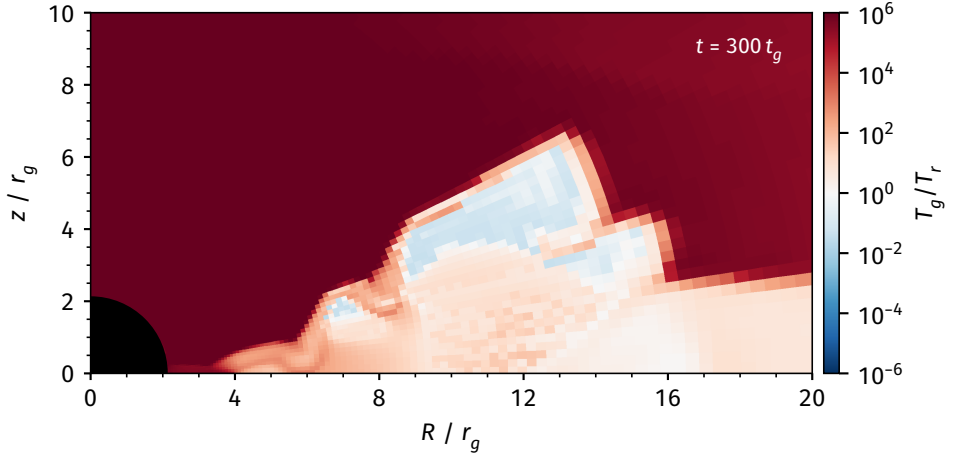
Early Evolution – Temperature Ratio



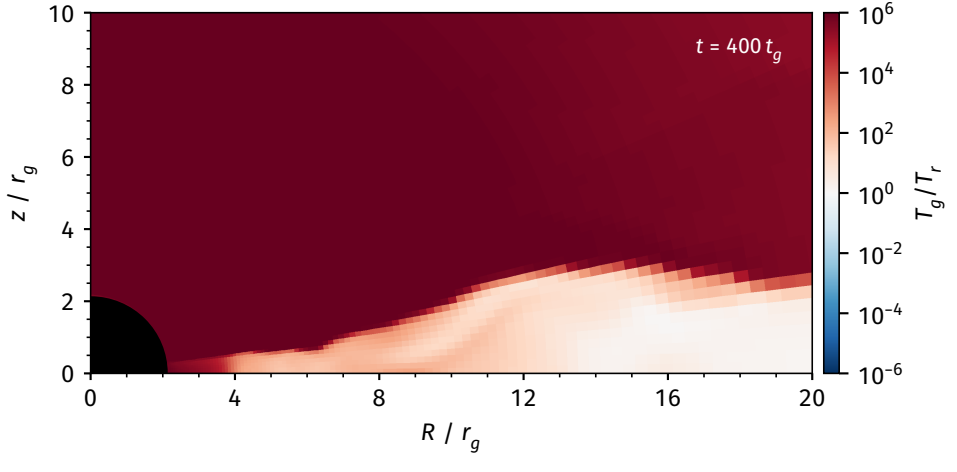
Early Evolution – Temperature Ratio



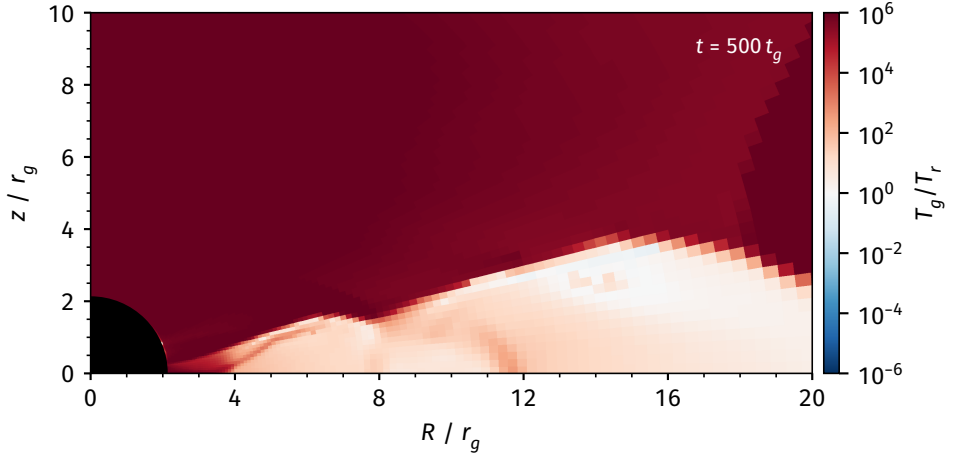
Early Evolution – Temperature Ratio



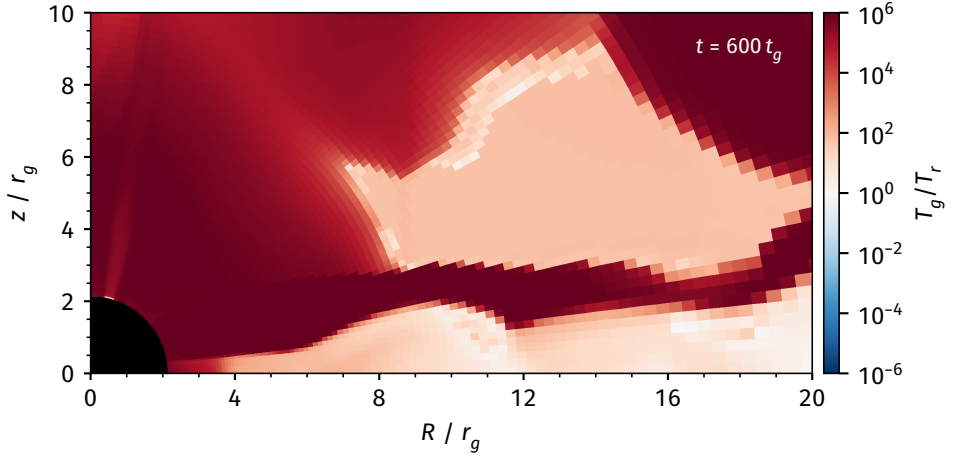
Early Evolution – Temperature Ratio



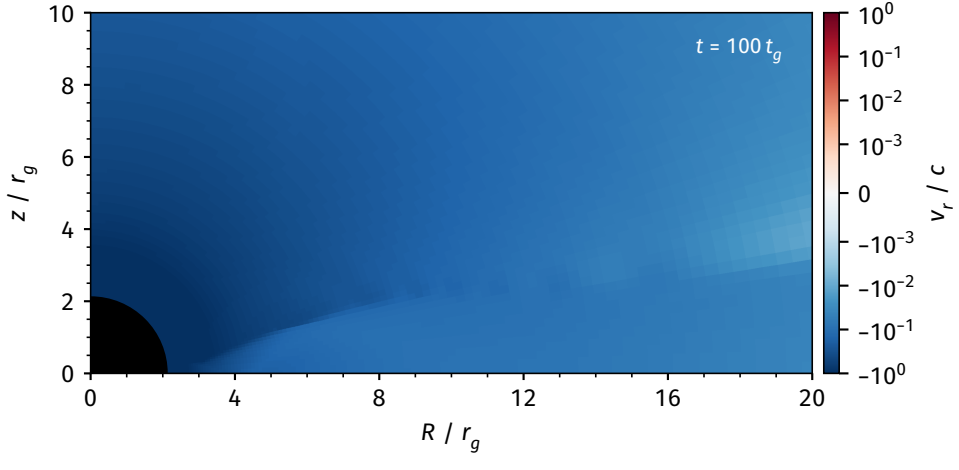
Early Evolution – Temperature Ratio



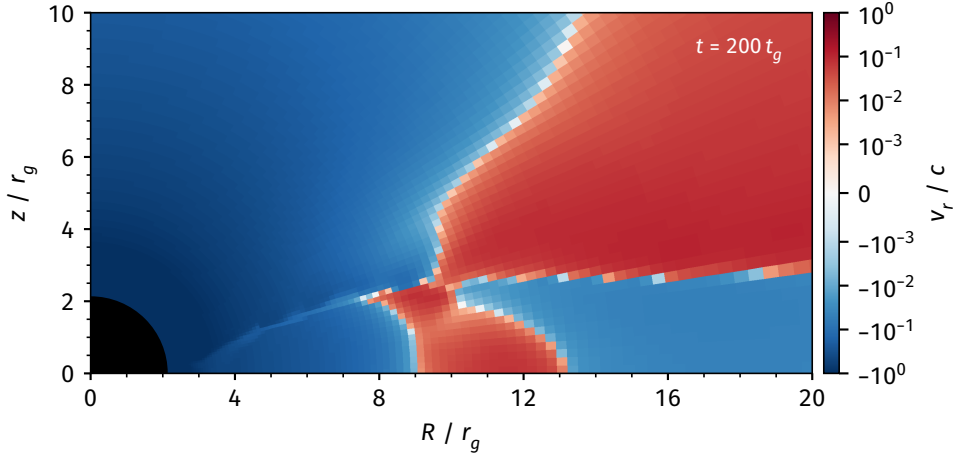
Early Evolution – Temperature Ratio



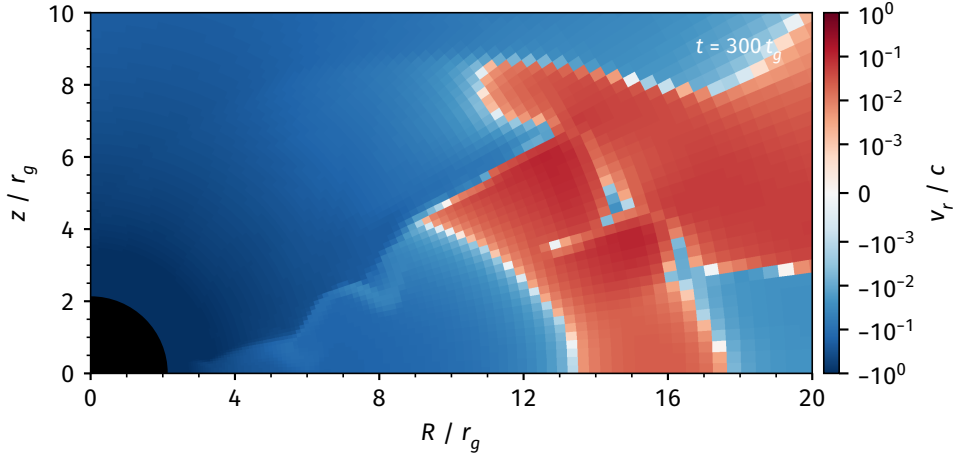
Early Evolution – Radial Velocity



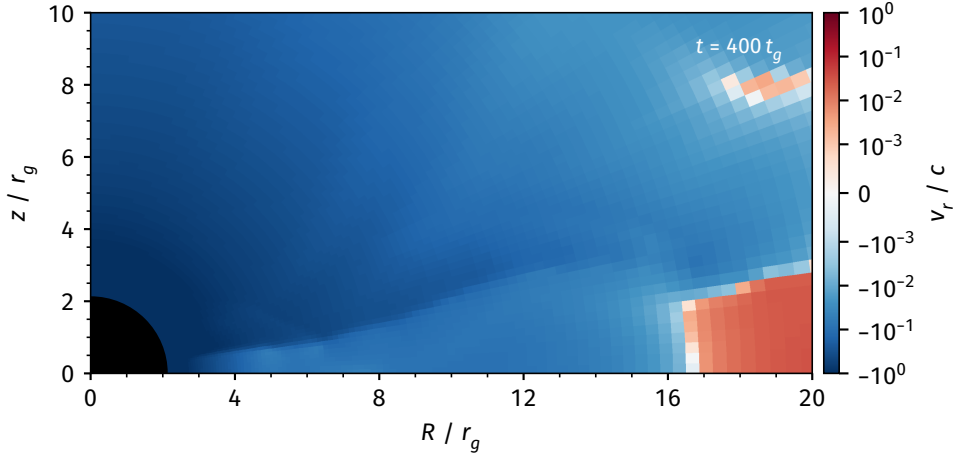
Early Evolution – Radial Velocity



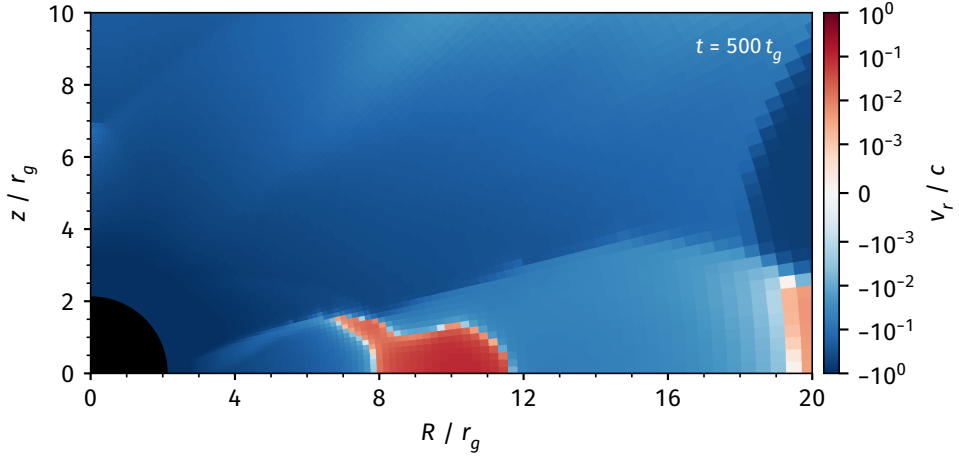
Early Evolution – Radial Velocity



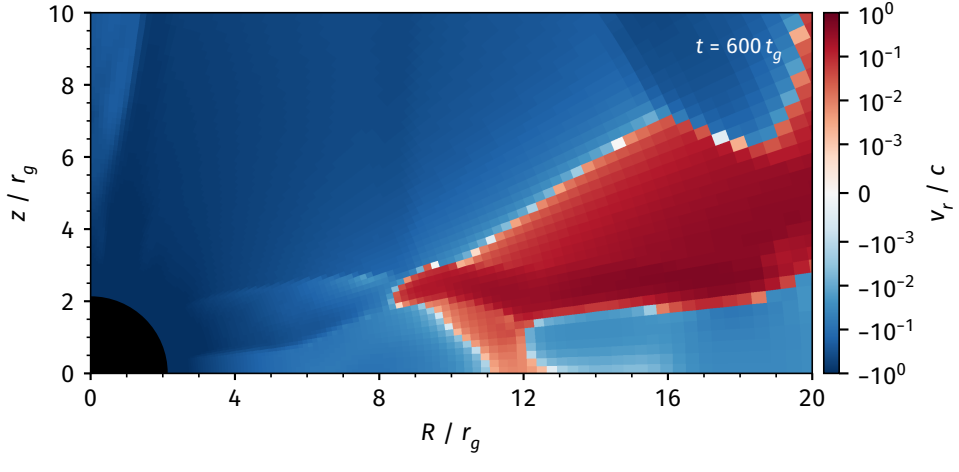
Early Evolution – Radial Velocity



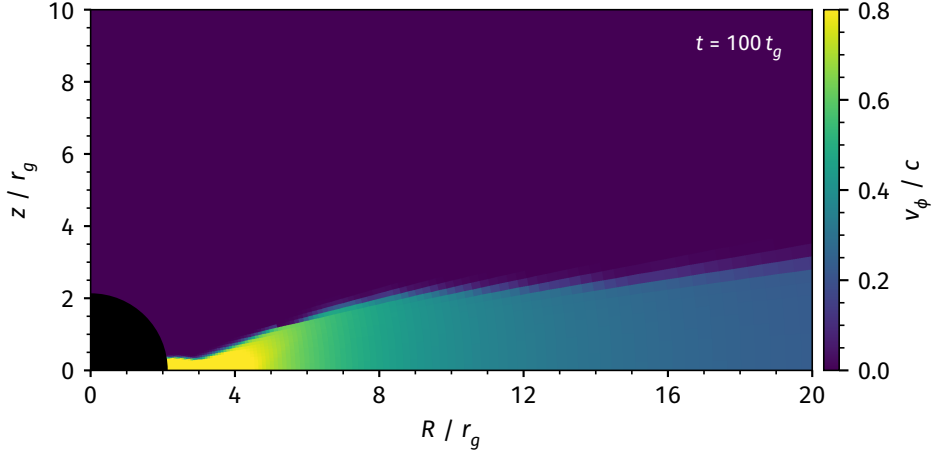
Early Evolution – Radial Velocity



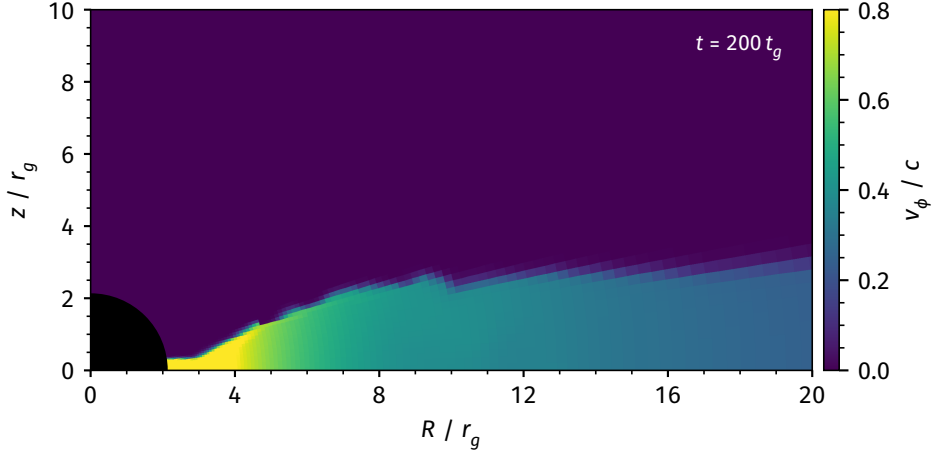
Early Evolution – Radial Velocity



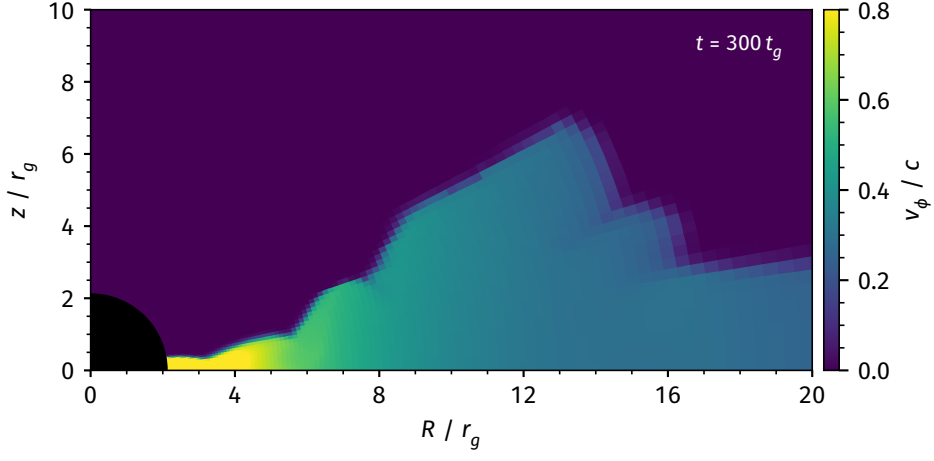
Early Evolution – Rotation Velocity



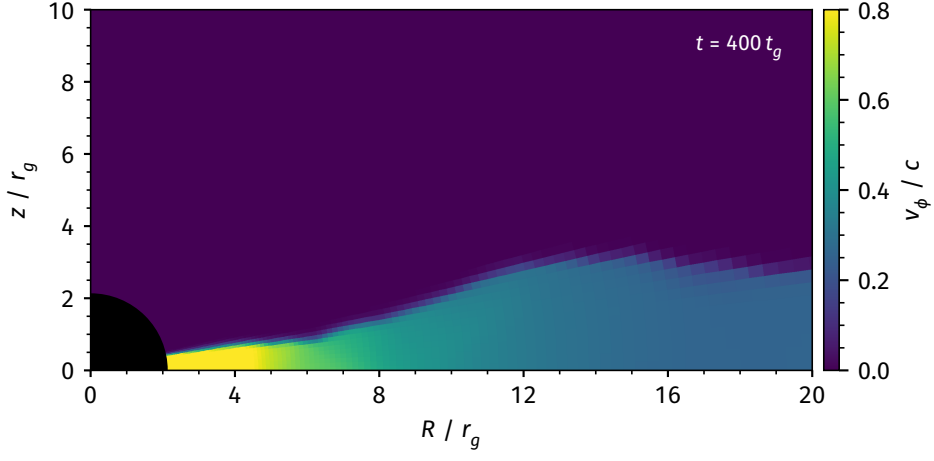
Early Evolution – Rotation Velocity



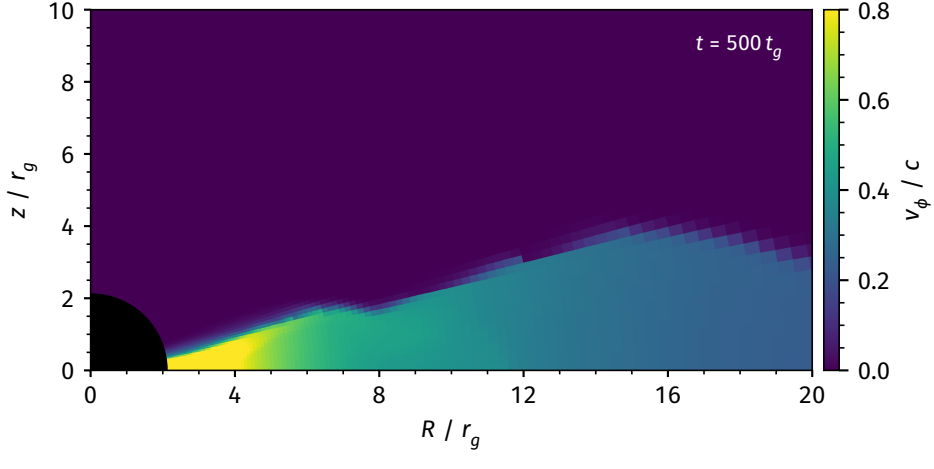
Early Evolution – Rotation Velocity



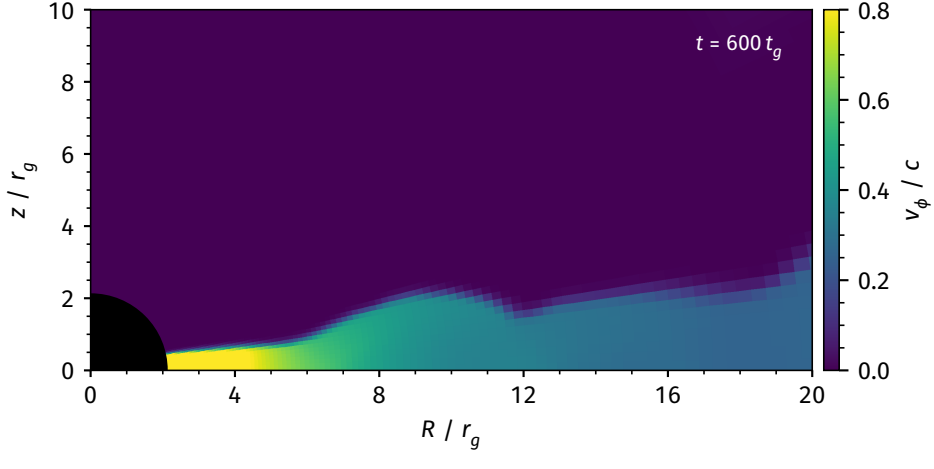
Early Evolution – Rotation Velocity



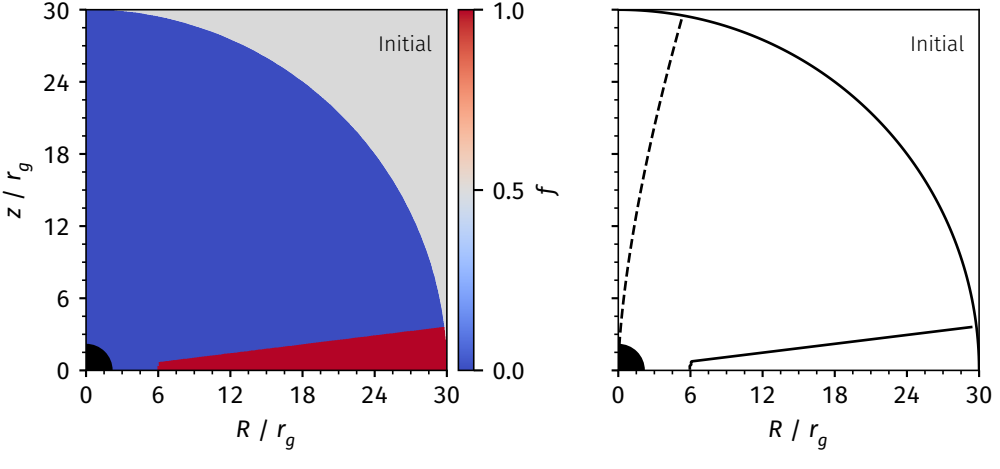
Early Evolution – Rotation Velocity



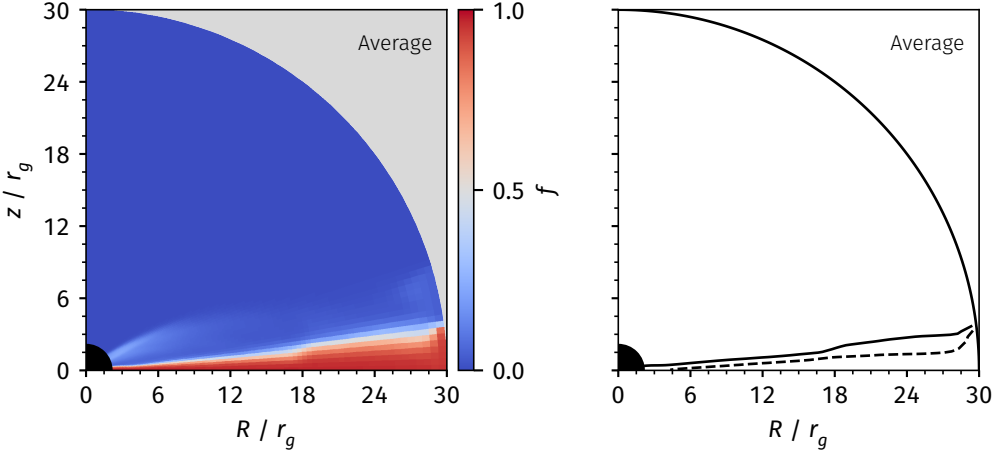
Early Evolution – Rotation Velocity



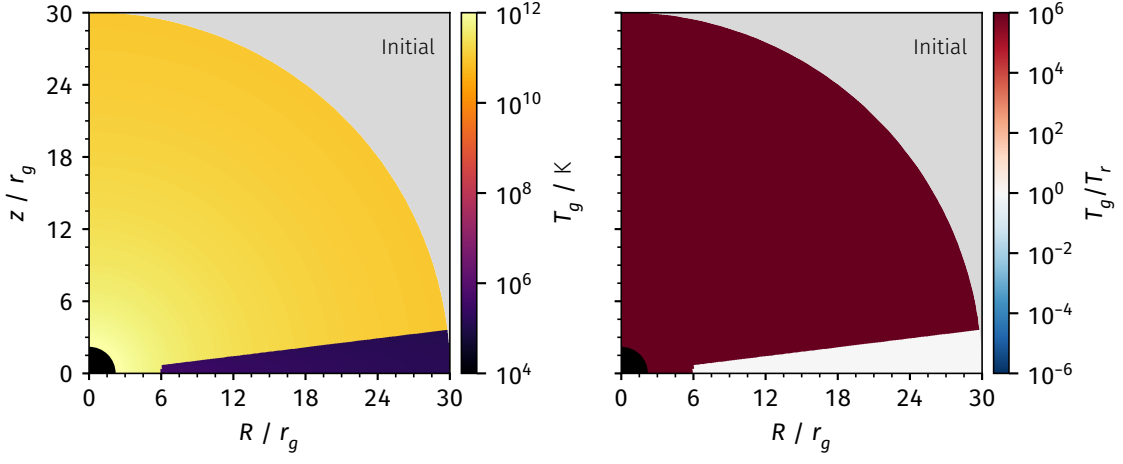
Steady State – DiskFraction & Disk Surface



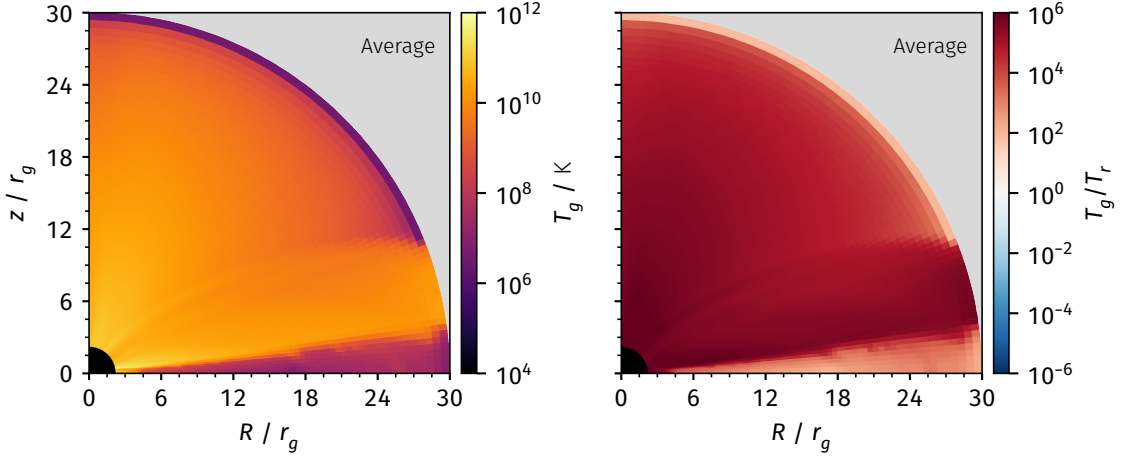
Steady State – DiskFraction & Disk Surface



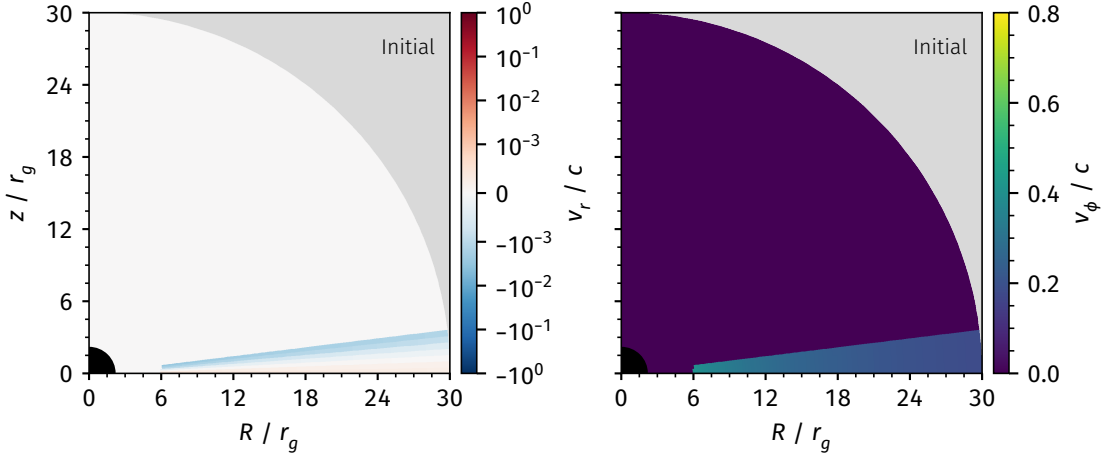
Steady State – Temperatures



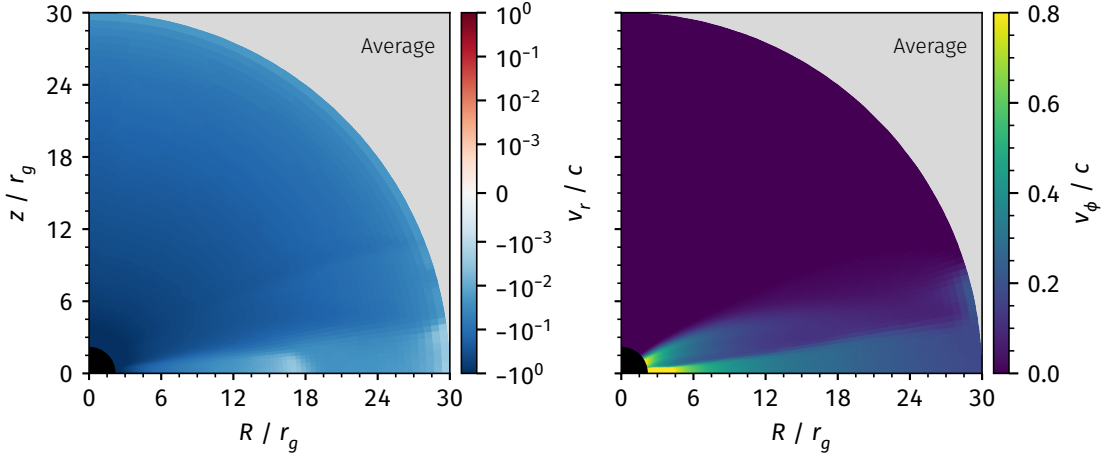
Steady State – Temperatures



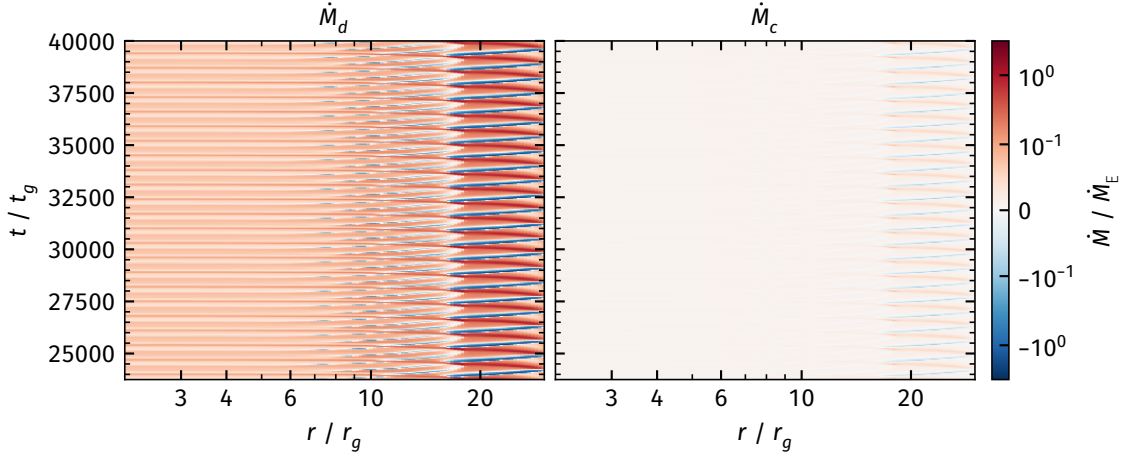
Steady State – Velocities



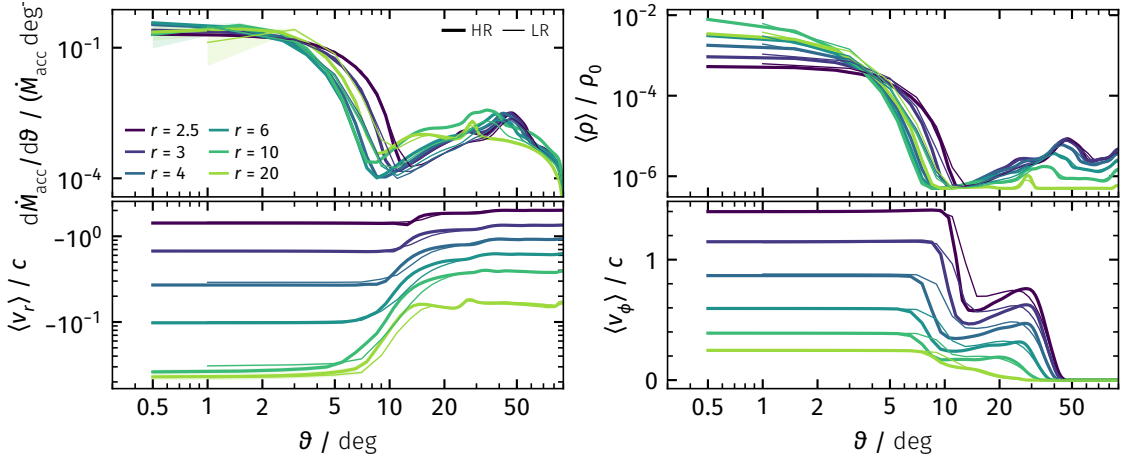
Steady State – Velocities



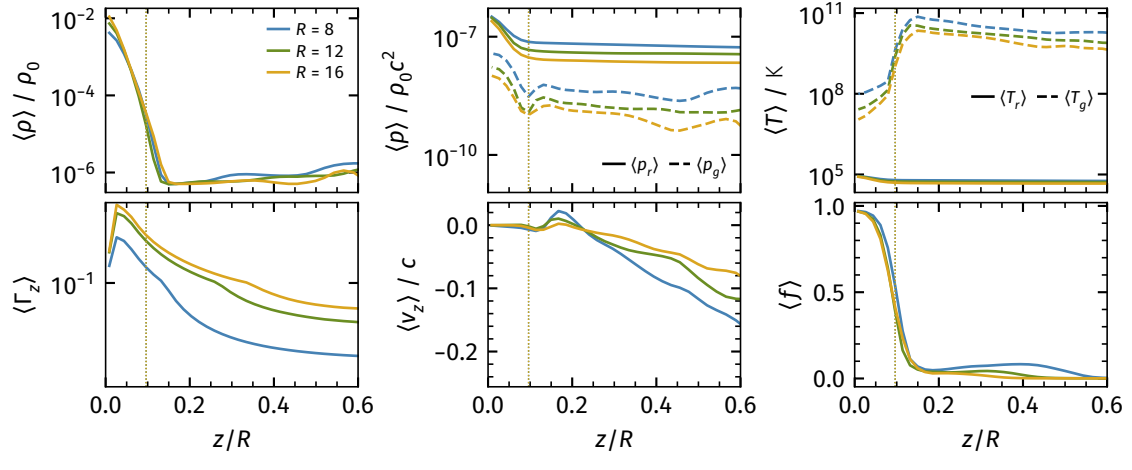
Steady State – Accretion Rate Map



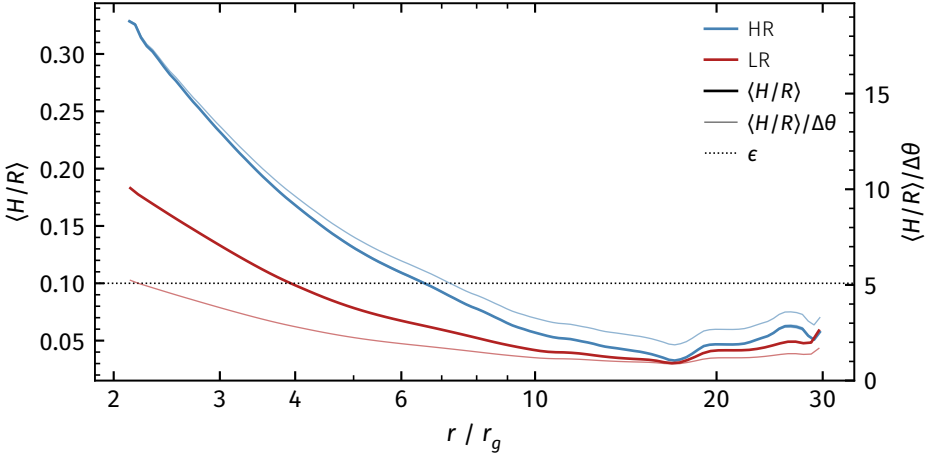
Steady State – Angular Profiles



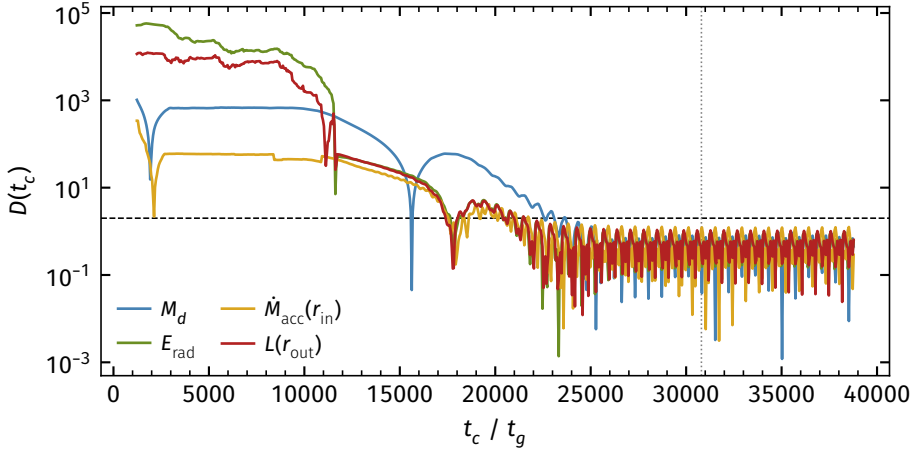
Steady State – Vertical Profiles



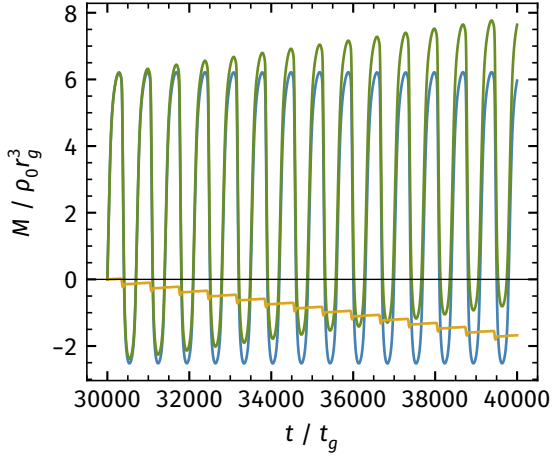
Steady State – Aspect Ratio

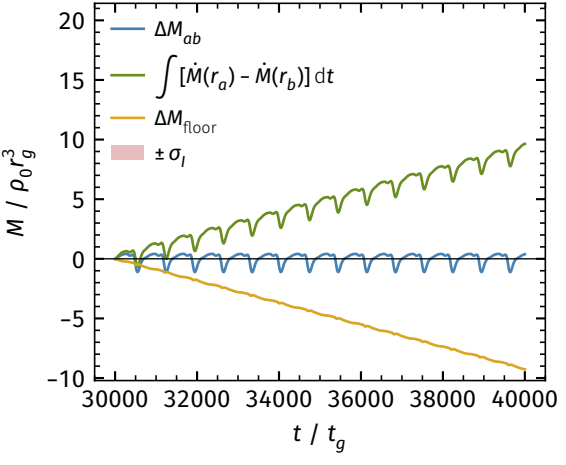


Diagnostics – Relaxation Statistic

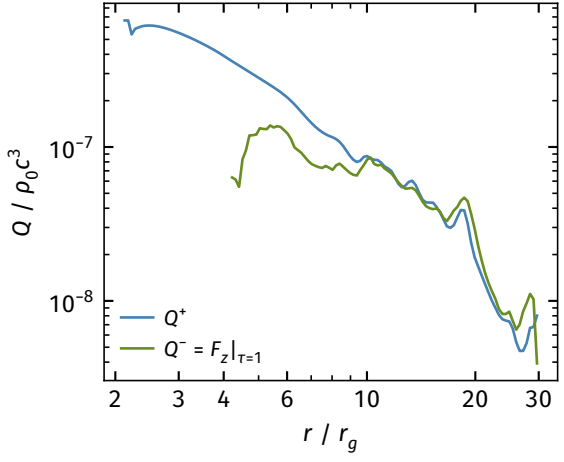


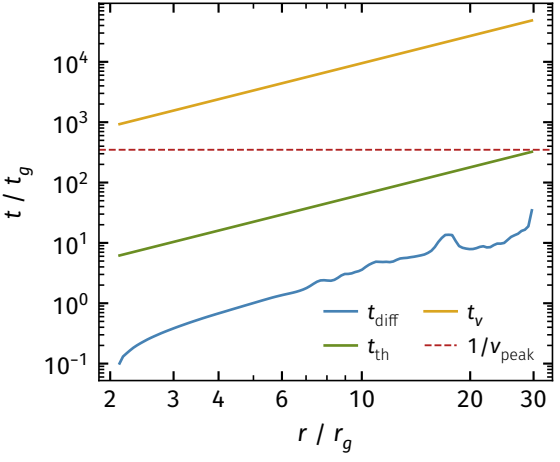
Diagnostics – Mass Budget



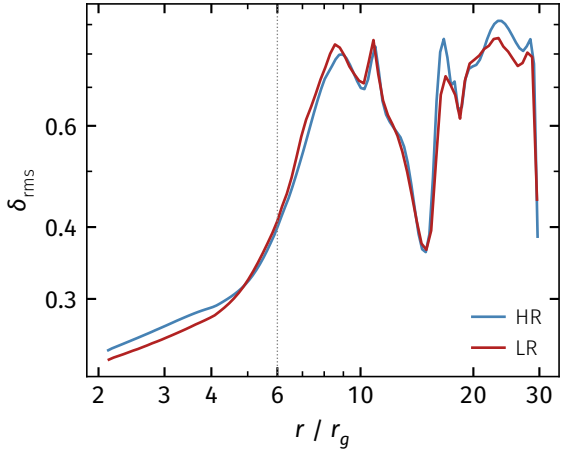


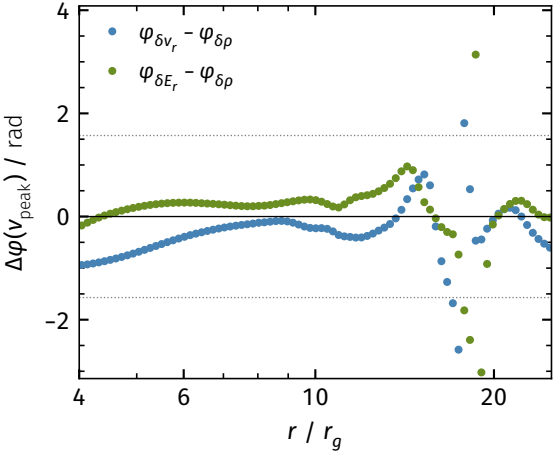
Diagnostics – Energy Balance





Diagnostics – Wave Amplitude & Cross Phase





Diagnostics – Timescales

